

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

**Gradient-Routed Alignment
with Capacity-aware Experts
for Multimodal Understanding**

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

HÀ NỘI - 2026

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Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Giáo sư Võ Đình Hiếu

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VIETNAM NATIONAL UNIVERSITY, HANOI
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**Gradient-Routed Alignment
with Capacity-aware Experts
for Multimodal Understanding**

BACHELOR'S THESIS

Major: Computer Science

Advisor: Dr. Nguyen Van Son

Co-Supervisor: Professor Vo Dinh Hieu

HANOI - 2026

Lời cảm ơn

Assurance

I hereby affirm that this thesis is my own original work. I have not submitted it previously for any academic or professional award.

I completed this thesis through my own efforts, with important guidance and support from Dr. Nguyen Van Son and Professor Vo Dinh Hieu for graduation purposes. I confirm that all information, data and findings presented here are original and free from plagiarism.

I have acknowledged every source that I consulted and included each one in the references. I accept full responsibility for the accuracy and integrity of this thesis under the regulations of VNU University of Engineering and Technology.

Hanoi, Ngày 16 tháng 4 năm 2026

Student

Le Xuan An

Tóm tắt

Khóa luận này nghiên cứu chứng cất tri thức đa giáo viên không đồng nhất cho các mô hình thị giác-ngôn ngữ (VLM) nhỏ gọn. Bài toán khó vì các teacher đến từ nhiều họ mô hình khác nhau, dùng tokenizer khác nhau, và không teacher nào hữu ích như nhau trên mọi mẫu. Vì vậy, cách lấy trung bình cố định giữa các teacher thường kém hiệu quả, đặc biệt với các student dưới một tỷ tham số.

Khóa luận đề xuất Gradient-Routed Alignment with Capacity-aware Experts (GRACE). Khung này xử lý trực tiếp ba vấn đề thực tế: lệch tokenizer, khác biệt thể mạnh giữa các teacher và bất ổn sau bước routing. Phương pháp kết hợp hàm mất mát trie-Wasserstein trên cây byte dùng chung, teacher gate phụ thuộc vào student, sparse top- k routing có ràng buộc capacity và cơ chế tinh chỉnh trọng số GRACE dựa trên gradient. Gate trước hết chọn một tập teacher nhỏ cho từng mẫu. Sau đó, GRACE so sánh gradient KD của từng teacher đã được chọn với gradient CE có giám sát của student, làm mượt điểm số bằng exponential moving average rồi chuyển các điểm này thành trọng số KD cuối cùng.

Thực nghiệm trên TextVQA, DocVQA và ChartQA cho thấy GRACE luôn tốt hơn finetuning trực tiếp và các baseline chứng cất đã so sánh. Trong thiết lập bốn teacher được báo cáo, GRACE đạt 69.38, 73.11 và 80.20 với SmolVLM-500M, và 60.73, 60.48 và 72.32 với SmolVLM-256M. Với SmolVLM-500M, mức cải thiện tương ứng so với mô hình gốc là 22.26%, 22.83% và 27.71%. Phương pháp cũng thu hẹp 55.03% khoảng cách tới teacher trên TextVQA, 39.53% trên DocVQA và 55.34% trên ChartQA.

Từ khóa: mô hình thị giác-ngôn ngữ, chứng cất tri thức, đa giáo viên, mixture-of-experts routing, chứng cất Wasserstein khác tokenizer

Abstract

This thesis addresses heterogeneous multi-teacher distillation for compact visual language models (VLMs). The setting is difficult because the teachers come from different model families, use different tokenizers and help different samples in different ways. A fixed teacher average is not enough, especially for sub-billion students.

We present GRACE, short for Gradient-Routed Alignment with Capacity-aware Experts. The framework directly addresses three practical failures at once: tokenizer mismatch, heterogeneous teacher specialization and unstable post-routing weighting. It combines a trie-Wasserstein KD loss on a shared byte trie, a student-conditioned teacher gate, sparse top- k capacity-aware routing and GRACE score refinement. The gate first selects a small teacher subset for each sample. GRACE then compares each routed teacher’s KD gradient with the student’s supervised CE gradient, smooths the agreement scores with an exponential moving average and converts them into final KD weights.

Experiments on TextVQA, DocVQA and ChartQA demonstrate that GRACE consistently improves over direct finetuning and the compared distillation baselines. In the reported four-teacher setting, GRACE reaches 69.38, 73.11 and 80.20 on SmolVLM-500M and 60.73, 60.48 and 72.32 on SmolVLM-256M. For SmolVLM-500M, this corresponds to gains of 22.26%, 22.83% and 27.71% over the original student. The method also closes 55.03% of the original teacher gap on TextVQA, 39.53% on DocVQA and 55.34% on ChartQA. These results establish GRACE as a strong distillation framework for compact text-rich VLMs.

Keywords: visual language model, knowledge distillation, multi-teacher distillation, mixture-of-experts routing, cross-tokenizer Wasserstein distillation

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List of Abbreviations

Abbr.	Full Term	Description
ANLS	Average Normalized Levenshtein Similarity	Evaluation metric used for DocVQA.
API	Application Programming Interface	Interface used to query model outputs in black-box settings.
bf16	Brain Floating Point 16	Low-precision floating-point format used for efficient mixed-precision training.
BLIP-2	Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models	A modular vision-language model with a Q-Former bridge.
CCA	Canonical Correlation Analysis	Classical similarity analysis method related to the interpretation of CKA scores.
CE	Cross-Entropy	Supervised loss used for classification or distillation objectives.
CKA	Centered Kernel Alignment	Representation similarity measure used to compare hidden layers between teacher and student models.
ChartQA	Chart Question Answering	Benchmark for question answering over charts.
CLIP	Contrastive Language-Image Pre-training	Dual-encoder vision-language pretraining framework.
CNN	Convolutional Neural Network	Standard visual backbone architecture used as a comparison point to ViT.
DocVQA	Document Visual Question Answering	Benchmark for question answering on document images.
EOS	End of Sequence	Special token that marks the end of a generated answer span.

Abbr.	Full Term	Description
FFN	Feedforward Network	Transformer sublayer applied after attention.
fp16	Floating Point 16	Half-precision floating-point format used in mixed-precision training or inference.
GRACE	Gradient-Routed Alignment with Capacity-aware Experts	Proposed multi-teacher distillation method in this thesis.
GPU	Graphics Processing Unit	Parallel hardware accelerator used for training and serving deep learning models.
SHM-CKA	Hidden-State Matching with CKA	Baseline distillation method that aligns teacher and student hidden representations using CKA.
InfoNCE	Information Noise-Contrastive Estimation	Contrastive objective for aligning image and text representations.
JSON	JavaScript Object Notation	Lightweight structured data format used for the LLaVA-style training files in this thesis.
JS	Jensen-Shannon	Symmetric divergence measure between probability distributions.
KD	Knowledge Distillation	Transfer of knowledge from a teacher model to a student model.
KL	Kullback-Leibler	Measure used to compare probability distributions.
KLD	Kullback-Leibler Divergence	Divergence-based objective widely used in distillation.
LLaVA	Large Language and Vision Assistant	Instruction-tuned projector-based vision-language model.
LLM	Large Language Model	Language model with large parameter scale and strong reasoning ability.
LN	Layer Normalization	Normalization method used to stabilize Transformer training.
MLP	Multi-Layer Perceptron	Feedforward projection block used in Transformers and classifiers.

Abbr.	Full Term	Description
MMD	Maximum Mean Discrepancy	Similarity metric used for feature-level alignment.
MoE	Mixture of Experts	Sparse architecture that routes tokens to selected experts.
MSA	Multi-Head Self-Attention	Attention block that models token interactions in Transformers.
MTKD	Multi-Teacher Knowledge Distillation	Distillation setting that uses more than one teacher model.
NLP	Natural Language Processing	Field concerned with computational understanding of language.
NLS	Normalized Levenshtein Similarity	String similarity score used in DocVQA evaluation.
OCR	Optical Character Recognition	Extraction of text from images or scanned documents.
OT	Optimal Transport	Distribution-matching perspective used to motivate tokenizer-agnostic distillation.
TWD	Tree-Wasserstein Distance	Tree-structured optimal-transport distance used as background for the trie-Wasserstein loss in this thesis.
PE	Positional Encoding	Representation of token order in Transformer models.
PLN	Parametric Layer Normalization	Layer normalization variant with trainable scale and bias.
Q-Former	Querying Transformer	Bridging module used by BLIP-2 to connect vision and language features.
QA	Question Answering	Task of generating answers to natural language questions.
ReLU	Rectified Linear Unit	Common non-linear activation function.
RMSNorm	Root Mean Square Layer Normalization	Lightweight normalization method based on root mean square statistics.
RoPE	Rotary Positional Embedding	Positional encoding method based on rotational transformations.

Abbr.	Full Term	Description
TF32	TensorFloat-32	NVIDIA tensor-core numeric format used to accelerate matrix operations during training.
TextVQA	Text Visual Question Answering	Benchmark for question answering on natural images containing text.
ULD	Universal Logit Distillation	Tokenizer-agnostic distillation loss based on ranked probability distributions.
ViT	Vision Transformer	Transformer-based model for image representation learning.
VLM	Vision-Language Model	Model that jointly processes visual and textual inputs.
VLM-Eval-Kit	Vision-Language Model Evaluation Kit	Toolkit used in this thesis to run standardized multimodal evaluation.
VNU	Vietnam National University	Parent university system of VNU University of Engineering and Technology.
VQA	Visual Question Answering	Task of answering questions grounded in visual inputs.
ZeRO	Zero Redundancy Optimizer	Memory-optimization technique used by DeepSpeed to reduce optimizer-state replication during large-model training.

Chapter 1

Introduction

1.1 Background and Motivation

The core task in this thesis is to distill strong multimodal ability from large VLM teachers into **sub-billion** VLM students. The goal is to do so without losing performance on text-centric benchmarks. Vision-Language Models (VLMs) jointly process images and text for tasks such as image captioning, visual question answering, document understanding and instruction following [2, 17–20].

In this thesis, *multimodal understanding* refers to practical capabilities such as reading text in images, understanding the input question and performing basic reasoning over visual and textual evidence, not broad general intelligence [7–9]. Our focus is efficient transfer of this ability to compact students.

Figure 1.1 illustrates the basic vision-text pretraining pattern that underlies many modern VLM families.

However, high-performing VLMs are often large and expensive in memory, latency and deployment cost [17, 18, 21, 22]. This cost limits their use in resource-constrained settings.

This pressure is consistent with scaling-law results. Kaplan et al. show that model quality improves predictably with scale, while Hoffmann et al. show that compute-optimal training still requires jointly increasing model size and data tokens [23, 24]. Larger models therefore remain attractive for accuracy, but they do not become cheap to train or deploy.

The vision side makes this constraint sharper in VLMs. VLMs insert visual embeddings or visual tokens into the multimodal context together with text, so the effective sequence length grows with image tokenization density. Since Transformer attention scales quadratically with sequence length, increasing the visual token budget quickly raises memory and latency [4, 19, 25]. For compact VLMs, this means that shrinking only the language backbone is not enough. The visual stack and the number of visual tokens also need to be controlled.

For this thesis, the most important regime is **small models**, especially **sub-billion VLMs**. Models in the 1B to 3B range are already cheaper than large 7B to 70B systems, but they can still be heavy for edge deployment, memory-limited serving, or repeated evaluation across many checkpoints. Sub-billion students are a stricter target. They force

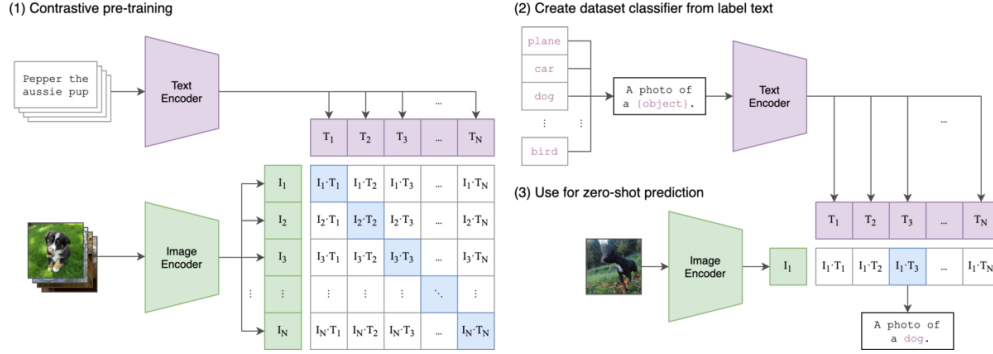


Figure 1.1: CLIP-style vision-language pretraining aligns image and text representations in a shared embedding space [1, 2]. This shared-space pretraining pattern is one of the main starting points for modern VLM families.

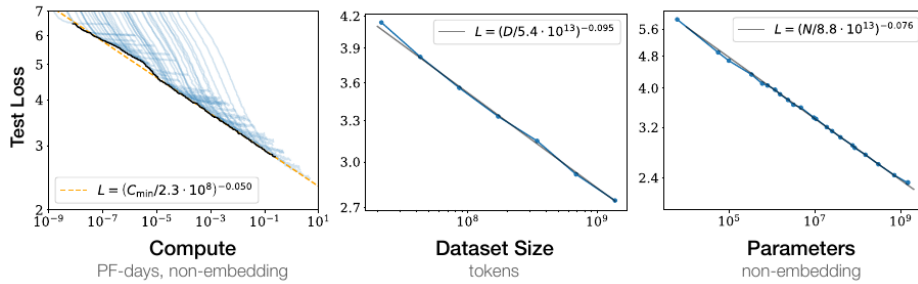


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

Figure 1.2: Scaling-law trends from Figure 1 of Kaplan et al. The reported loss decreases smoothly as compute, dataset size and parameter count increase, which explains why larger teachers remain attractive even when deployment favors smaller students [23].

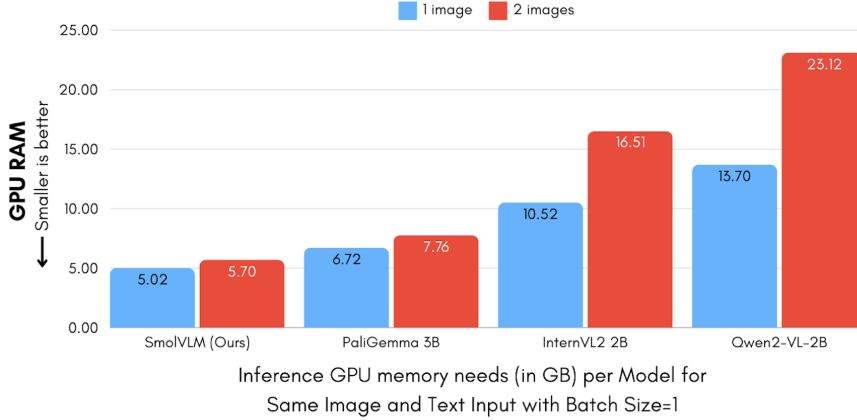


Figure 1.3: Inference graphics processing unit (GPU) memory usage reported in the SmolVLM release blog. The comparison shows that compact VLM design can reduce memory substantially relative to similarly capable models, especially when the visual token budget is controlled [25, 26].

the model to use its parameter budget and visual tokens more carefully, while still aiming to retain useful multimodal performance [21, 22, 25].

Figure 1.3 highlights the deployment side of the problem, while the next figure places compact VLM families on the parameter scale itself.

The issue is especially important for text-centric tasks such as scene-text reading, document understanding and chart reasoning [7–9].

Figures 1.2, 1.3, 1.4 and 1.5 together motivate the same conclusion: strong teachers benefit from scale, but the deployment target is still a much smaller student.

Knowledge distillation (KD) transfers knowledge from a large teacher to a compact student, often preserving strong performance at lower cost [11, 30].

Figure 1.5 shows why this direction matters. Compact multimodal models can already be competitive, but closing the gap to stronger models still requires better transfer of knowledge.

Figure 1.6 sharpens this point. On these three benchmarks, the best open-source models are already close to the best proprietary models and in two cases they are ahead.

This makes white-box distillation especially attractive. In black-box KD, the student can only query an external teacher service and read its returned predictions. In white-box KD, the student can also use teacher logits and intermediate states [31, 32]. That extra access enables objectives such as the trie-Wasserstein KD loss used in this thesis, as well as prior baselines such as ULD and hidden-state matching, which are not available in a pure black-box setting [12, 15, 30, 33].

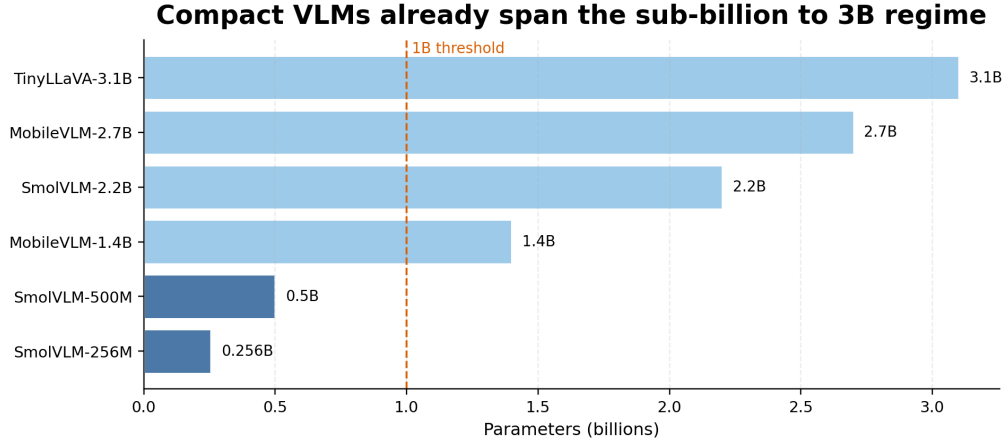


Figure 1.4: Representative compact VLM families already span the sub-billion to 3B regime. This is the design space most relevant to our thesis: the student must stay small enough for efficient deployment, while still benefiting from stronger teachers [21, 22, 25].

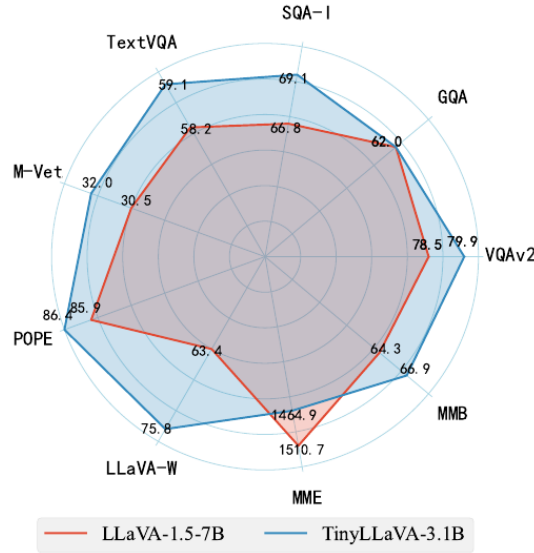


Figure 1.5: Benchmark comparison from TinyLLaVA Figure 1, where a 3.1B small-scale multi-modal model remains competitive with the 7B LLaVA-1.5 baseline across several benchmarks [22].

Open-source frontier models are already close to proprietary leaders

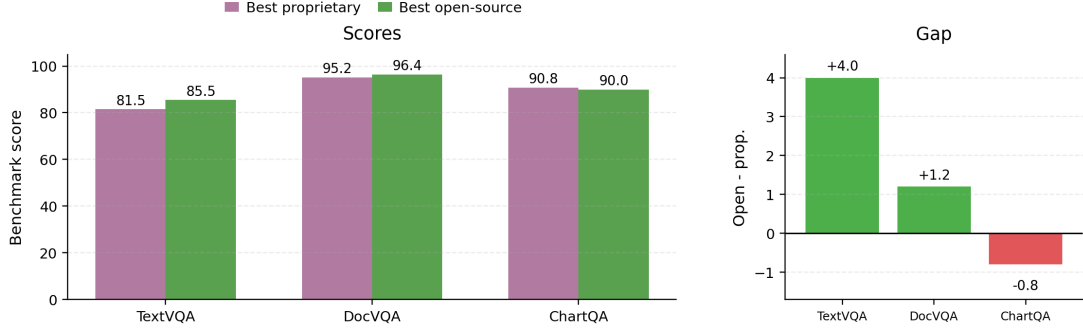


Figure 1.6: Representative proprietary and open-source frontier results on TextVQA, DocVQA and ChartQA. The gap is already small and the best open-source model is ahead on TextVQA and DocVQA. The right panel shows the benchmark gap computed as open-source score minus proprietary score from snapshots accessed on April 2, 2026 [27–29].

Another major competing compression path is quantization. Unlike distillation, quantization compresses an already trained model by reducing numerical precision. It can cut memory and improve inference speed substantially, so it is a serious practical alternative to KD [34–36].

In VLMs, however, naive low-bit quantization can hurt multimodal accuracy [36, 37]. For this reason, we treat quantization as a complementary deployment step, not a substitute for better transfer during training.

This observation is exactly why we focus on **sub-billion vision-language models**. They are small enough to make the efficiency constraint real, yet strong enough that improvements from better distillation remain meaningful [25].

In the multimodal setting, recent work has shown that distillation can transfer reasoning or representation knowledge into more efficient VLMs [20, 38, 39]. This is especially relevant on text-rich benchmarks such as TextVQA, DocVQA, and ChartQA [7–9].

Figure 1.8 summarizes the teacher-student transfer pattern that this thesis extends to the heterogeneous multi-teacher setting.

Since no single teacher excels across all multimodal tasks, *multi-teacher distillation* can provide richer supervision than a single-teacher pipeline [40–42].

In our setting, this is not just a matter of picking several strong teachers. The teacher pool contains heterogeneous VLMs with different tokenizers, different output vocabularies and different sample-level strengths [12, 40, 41].

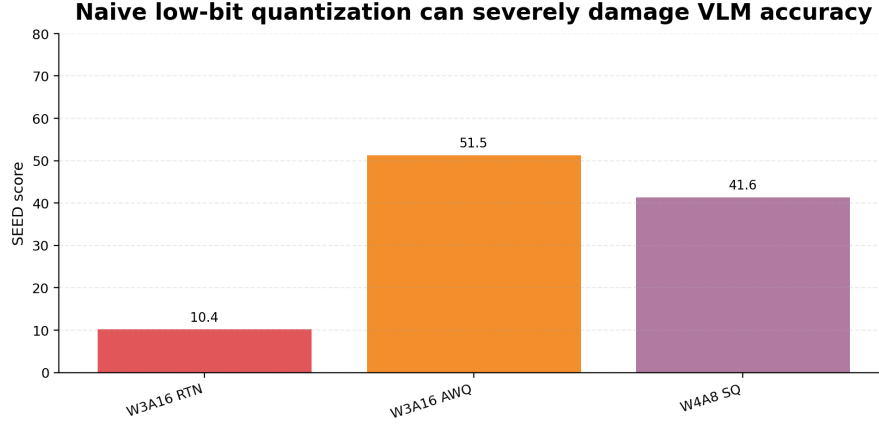


Figure 1.7: SEED accuracy for LLaVA-OneVision-7B under several low-bit quantization settings, reconstructed from MBQ Table 4 [37]. The naive W3A16 RTN baseline drops by 60.7 points, which is why we treat quantization as complementary to distillation, not a replacement for it.

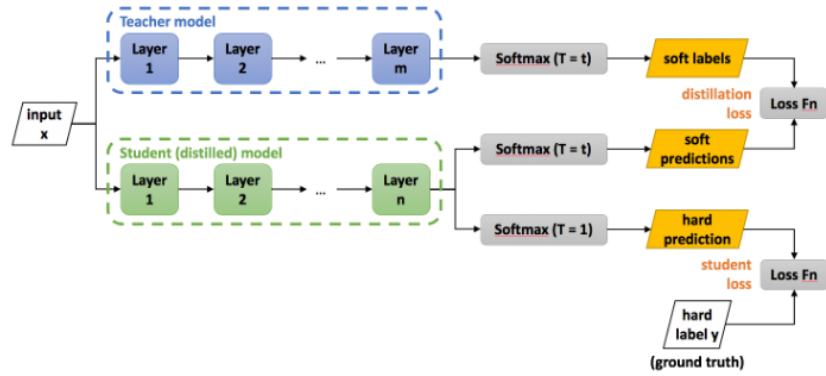


Figure 1.8: Knowledge distillation transfers information from a teacher model to a smaller student model [10, 11]. This thesis extends that basic pattern to heterogeneous teachers and sample-dependent weighting.

A naive uniform average can therefore mix teachers that help one sample but hurt another [30, 40, 41].

This motivates a selective multi-teacher framework over static teacher averaging.

We introduce Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a selective multi-teacher distillation framework that routes teachers per sample, applies a trie-Wasserstein cross-tokenizer loss and refines teacher weights with gradient-based alignment. The framework tackles tokenizer mismatch, heterogeneous teacher specialization and post-routing instability in one recipe. The next section formalizes the problem and requirements.

1.2 Problem Statement and Research Objectives

This thesis addresses one central problem: how to transfer multimodal understanding from large VLMs to compact students. The transfer must preserve task performance while cutting deployment cost.

We focus on text-rich multimodal tasks where the student must read image text, understand the question and perform basic reasoning [7–9].

In this setting, existing distillation pipelines often assume a single teacher, a shared tokenizer, or a fixed averaging rule across teachers. These assumptions are restrictive for heterogeneous multi-teacher VLM distillation [11, 12, 40, 41].

More specifically, the method should satisfy three requirements.

1. The KD loss should remain valid even when teacher and student tokenizers differ.
2. Teacher assignment should be *sample-dependent*, not fixed across the whole dataset.
3. The routing mechanism should remain sparse while suppressing routed teachers whose KD direction conflicts with the supervised objective. When needed, the framework should also support regularizers that discourage routing collapse.

Given a pool of N frozen teacher VLMs and a compact student, the goal is therefore to learn a distillation strategy that:

1. Remains valid across mismatched teacher and student vocabularies.
2. Routes teachers adaptively from student-side features through a learned student-conditioned gate.

3. Retains sparse top- k computation under capacity constraints.
4. Filters routed teachers when their KD update direction conflicts with the supervised objective, while keeping a fallback teacher when necessary.

This thesis addresses these requirements with Gradient-Routed Alignment with Capacity-aware Experts (GRACE).

GRACE is a selective multi-teacher distillation framework.

It combines a trie-Wasserstein cross-tokenizer KD loss, a student-conditioned teacher gate, top- k capacity-aware sparse routing and gradient-based GRACE score refinement. The framework also supports routing fallback after capacity pruning, uniform fallback when score spread is too small and balancing and expert-bias updates [12, 14, 43–45].

We evaluate it on text-rich multimodal understanding tasks such as TextVQA, DocVQA and ChartQA [7–9].

Accordingly, this thesis focuses on cross-tokenizer multimodal distillation for compact students and evaluates GRACE on TextVQA, DocVQA and ChartQA, with analysis of routing behavior and representation matching.

1.3 Thesis Contributions

The main contributions are:

1. We define a heterogeneous multi-teacher distillation setting for compact VLMs with mismatched teacher and student vocabularies on TextVQA, DocVQA and ChartQA.
2. We present Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a selective multi-teacher distillation framework that combines trie-Wasserstein cross-tokenizer KD, a student-conditioned gate, sparse top- k routing and GRACE score refinement.
3. We show that this framework directly addresses the main practical failures of heterogeneous VLM distillation: tokenizer mismatch, sample-dependent teacher conflict and unstable teacher weighting after routing.
4. We establish consistent gains over fine-tuning and strong distillation baselines, supported by routing, ablation and representation analyses.

Chapter 2

Preliminaries

2.1 Transformer-Based Vision-Language Models

2.1.1 Transformer Basics

Modern vision-language models are built on the Transformer architecture [4]. For an input sequence, self-attention maps query, key and value matrices $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ to

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}, \quad (2.1)$$

where d_k is the key dimension. Multi-head attention runs this operation in parallel across several learned subspaces, then applies a feed-forward network and residual normalization.

For language modeling, the Transformer produces a sequence of hidden states and next-token logits. In this thesis, the student and teacher interact through their output logits because the distillation method operates at the token-distribution level [4, 11].

2.1.2 Vision Encoder and Multimodal Fusion

On the visual side, many multimodal systems use a Vision Transformer (ViT) [6], which converts an image into a sequence of patch embeddings and processes them with Transformer blocks. A connector module then fuses these visual tokens with text, using cross-attention, a Q-Former, or a learned projector [17–19].

This thesis focuses on compact generative VLMs in the same general family as SmolVLM and related projector-based models. Conceptually, we can write such a model as

$$\mathbf{z} = f_\theta(x), \quad (2.2)$$

where x is a multimodal input and $\mathbf{z}$ denotes the output logits used for autoregressive answer generation.

Figures 2.1 and 2.2 give the basic Transformer and ViT blocks that later multimodal systems build on.

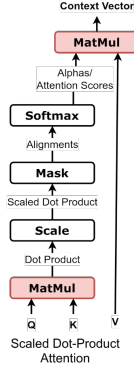


Figure 2.1: Transformer attention block structure [3, 4]. The figure isolates the attention and feed-forward block that is later reused in both language and vision stacks.

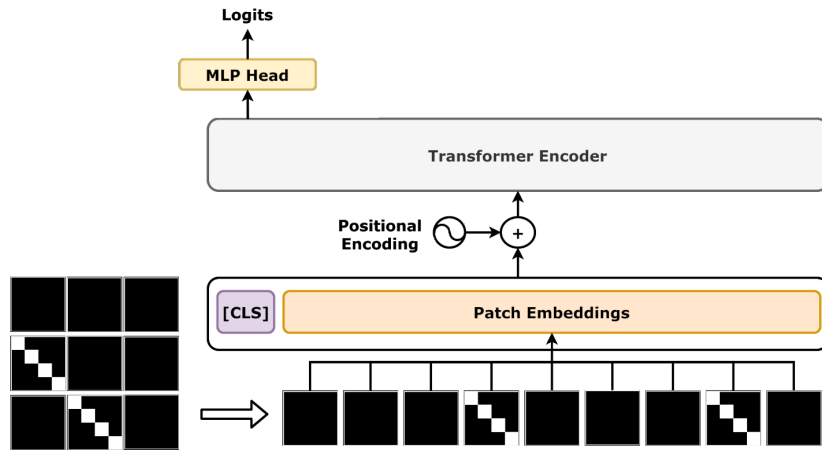


Figure 2.2: Vision Transformer pipeline with patch embedding and Transformer encoder blocks [5, 6]. It shows how an image becomes a token sequence before multimodal fusion.

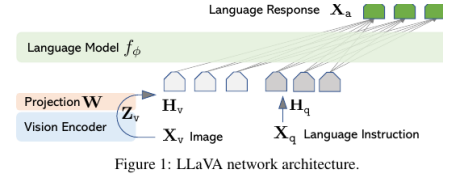
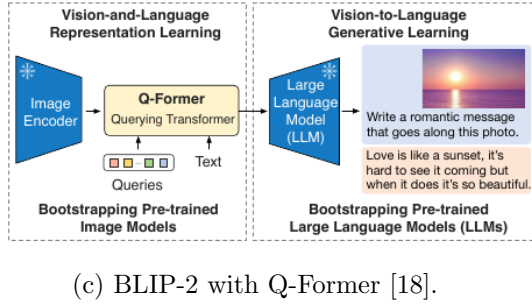
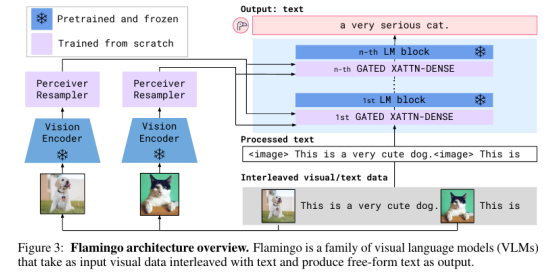
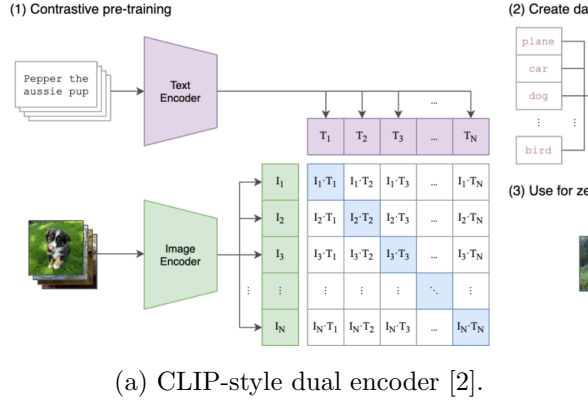


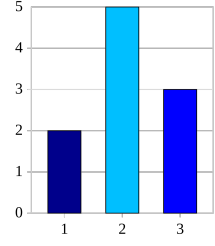
Figure 2.3: Representative architecture families of modern vision-language models. Together they show the main fusion choices used in practice: dual encoders, cross-attention, query bottlenecks and projector-based coupling to a language model.



(a) TextVQA



(b) DocVQA



(c) ChartQA

Figure 2.4: Representative inputs from the text-rich multimodal benchmarks considered in this thesis [7–9]. The three examples show why text reading, layout parsing and visual reasoning all matter in this benchmark suite.

2.1.3 Text-Rich Multimodal Understanding

The tasks considered in this thesis, namely TextVQA, DocVQA and ChartQA, require the model to read text in images, understand the question and perform basic reasoning over visual and textual evidence [7–9]. This setting differs from generic image captioning because success depends strongly on optical character recognition (OCR)-sensitive recognition and token-level answer generation.

Figure 2.3 summarizes the main architecture families, while Figure 2.4 shows the text-rich inputs used in this thesis.

2.2 Knowledge Distillation

2.2.1 Teacher-Student Learning

Knowledge distillation transfers information from a stronger teacher model to a smaller student model [11, 30]. Let $\mathcal{L}_{CE}$ denote the supervised cross-entropy loss of the student. A generic distillation objective takes the form

$$\mathcal{L} = \lambda_{CE}\mathcal{L}_{CE} + \lambda_{KD}\mathcal{L}_{KD}, \quad (2.3)$$

where $\mathcal{L}_{KD}$ is a teacher-guided loss. In later chapters, GRACE instantiates this template with a routed trie-Wasserstein term plus small router regularizers.

In token-level distillation, the teacher and student are compared on their output logits, not only on hard labels. This allows the student to learn richer distributional structure, often called *dark knowledge* [11].

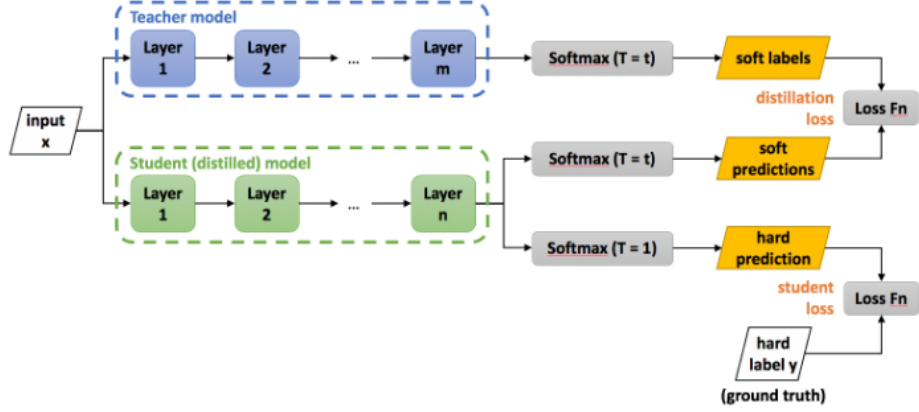


Figure 2.5: Generic teacher-student knowledge distillation setup with hard labels and soft targets [10, 11]. It provides the basic supervision pattern before GRACE adds heterogeneous teachers and routing.

Figure 2.5 provides the generic teacher-student template before Chapter 3 specializes it to GRACE.

2.2.2 Temperature-Scaled Output Distributions

Let $\mathbf{z}_s$ and $\mathbf{z}_t$ denote student and teacher logits. With temperatures τ_s and τ_t , the corresponding distributions are

$$\mathbf{p}_s = \text{softmax}\left(\frac{\mathbf{z}_s}{\tau_s}\right), \quad \mathbf{p}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau_t}\right). \quad (2.4)$$

Many classical KD methods compare $\mathbf{p}_s$ and $\mathbf{p}_t$ using divergences such as forward KL or Jensen-Shannon divergence. However, such approaches usually assume that teacher and student share the same vocabulary [11, 12, 46].

2.3 Cross-Tokenizer Distillation Background

2.3.1 Motivation

In this thesis, the teacher pool may contain different VLM families with different tokenizers and vocabulary sizes. A KD loss that assumes a shared output space is therefore too restrictive. Universal Logit Distillation (ULD) is one important prior remedy because it avoids direct token-ID matching [12]. Chapter 3 later replaces the ranked-mass surrogate with a trie-Wasserstein variant that keeps the same cross-tokenizer goal but uses a tree-structured transport view.

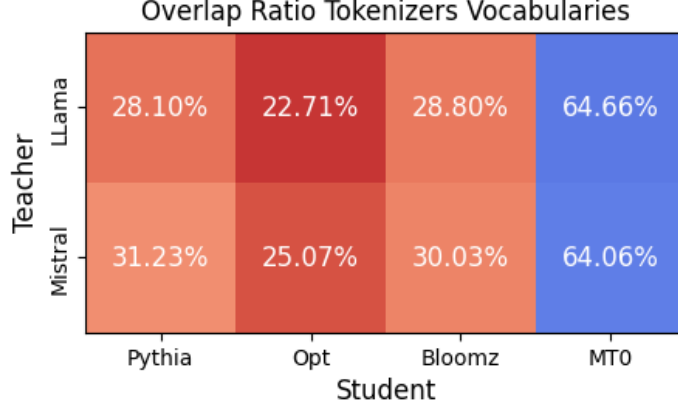


Figure 2.6: Vocabulary-overlap ratios between teacher and student tokenizers from the ULD paper [12]. The overlap is far from complete, which makes direct token-ID KD unreliable across model families.

Figure 2.6 makes the mismatch explicit: even strong teacher-student pairs can share only part of their vocabularies.

2.3.2 ULD as a Prior Baseline

Given student logits $\mathbf{z}_s \in \mathbb{R}^{V_s}$ and teacher logits $\mathbf{z}_t \in \mathbb{R}^{V_t}$, we first compute their temperature-scaled probability distributions $\mathbf{p}_s$ and $\mathbf{p}_t$. We then sort both vectors in descending order, pad the smaller vocabulary with zeros and compare the sorted masses with an L_1 distance:

$$\mathcal{L}_{ULD}(\mathbf{z}_s, \mathbf{z}_t) = \|\pi(\mathbf{p}_s) - \pi(\mathbf{p}_t)\|_1, \quad (2.5)$$

where $\pi(\cdot)$ denotes sorting in descending order after zero-padding.

The key advantage of ULD is that it compares distributions by ranked probability mass instead of token identity. This makes it directly applicable to cross-tokenizer teacher-student distillation [12]. In the later experiments, ULD is treated as a cross-tokenizer baseline, not the final method.

2.3.3 Optimal Transport and Tree-Structured Mass Matching

Optimal transport compares two probability measures by the minimum cost required to move mass from one distribution to the other [14]. For discrete source weights $\mathbf{a}$, target weights $\mathbf{b}$ and transport cost matrix $\mathbf{C}$, the corresponding optimization problem can be written as

$$W(\mathbf{a}, \mathbf{b}) = \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{P}, \mathbf{C} \rangle, \quad \mathcal{U}(\mathbf{a}, \mathbf{b}) = \{\mathbf{P} \geq 0 \mid \mathbf{P}\mathbf{1} = \mathbf{a}, \mathbf{P}^\top \mathbf{1} = \mathbf{b}\}. \quad (2.6)$$

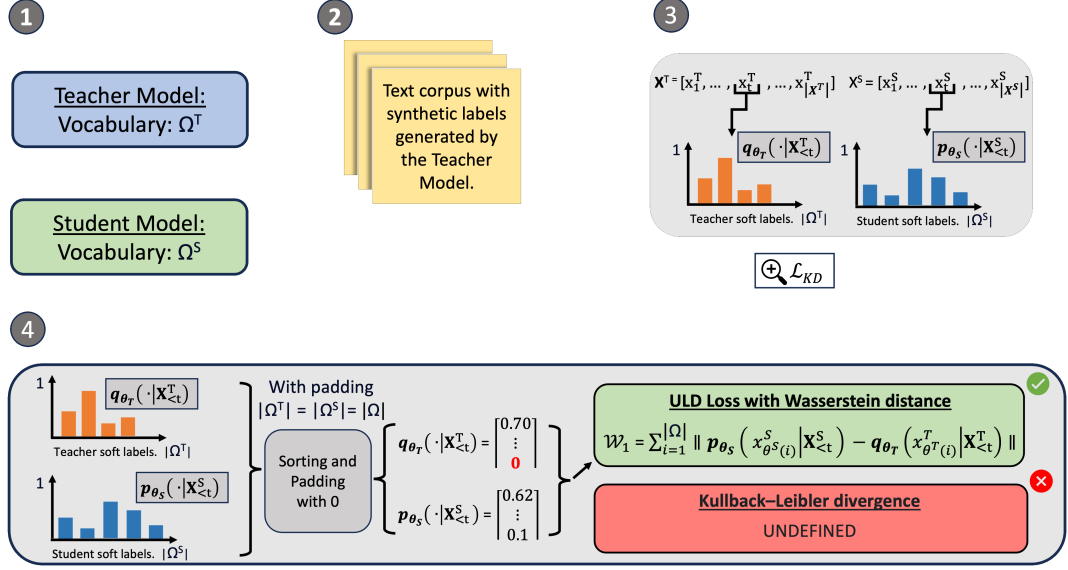


Figure 2.7: ULD pipeline from the original paper [12]. Teacher and student probabilities are sorted and zero-padded before the ranked masses are compared, which removes the need for direct vocabulary alignment.

Here, $\mathbf{P}$ is a transport plan that specifies how much mass is moved from each source bin to each target bin.

Figure 2.8 gives the continuous geometric view of OT. The same idea appears in the discrete case through a transport plan $\mathbf{P}$, and later in the tree case through edgewise subtree mass.

This viewpoint also motivates tree-structured transport. When teacher and student have different vocabularies, matching probabilities by token index is not meaningful. What matters instead is how probability mass sits on a shared structure [12, 14].

Tree-Wasserstein distances make that idea explicit by placing probability mass on a tree metric and summing weighted subtree imbalances [14, 15]. For a rooted tree $T = (V, E)$ with edge weights $w(e)$, one convenient form is

$$W_T(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|, \quad (2.7)$$

where $\Gamma(e)$ denotes the subtree below edge e . The key point is that, on a tree metric, transport decomposes into edgewise subtree-mass imbalance instead of a dense source-target coupling [15, 47]. This is computationally attractive because the comparison follows the tree structure directly and it is conceptually natural when tokens share prefixes or other hierarchical relations.

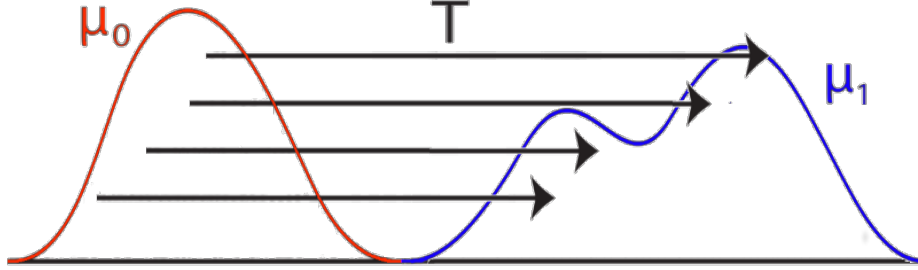


Figure 2.8: Continuous displacement view of optimal transport [13, 14]. The figure emphasizes the geometric side of OT: mass is moved through space with a transport map, not compared by fixed bin identity alone.

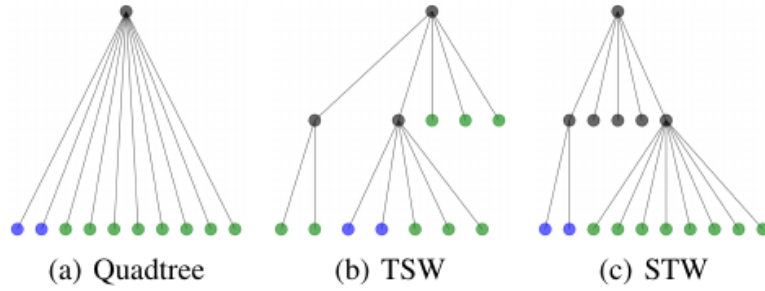


Figure 2.9: Example tree metrics from Quadtree, tree-sliced Wasserstein (TSW) and supervised tree-Wasserstein (STW) on a synthetic dataset [15]. The comparison is not a VLM result, but it makes the tree-transport idea explicit: mass is organized along branches of a hierarchy instead of through one dense transport matrix.

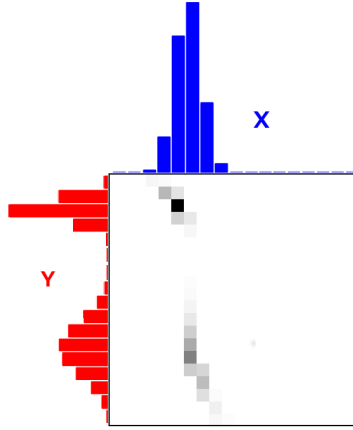


Figure 2.10: Discrete transport plan matrix in optimal transport, showing how source and target marginals are coupled by a transport matrix [14, 16]. The key idea is geometric: mass is compared by how it can be rearranged, not by fixed token identity.

Tree-sliced OT extends the same idea by averaging Wasserstein distances over tree metrics and shows that tree metrics preserve the geometric view of OT while admitting fast closed-form computations [47]. In this thesis, the relevant tree is not an abstract hierarchy but the shared UTF-8 byte trie induced by teacher and student decoded single-token strings. That perspective is closer to the trie-Wasserstein loss used later in Chapter 3: the method maps tokens to paths on a shared byte trie and compares mass on edges instead of token index.

Figures 2.8, 2.10 and 2.9 now cover the three views used later in the thesis: geometric mass movement, discrete transport plans and tree-structured mass matching. Chapter 3 instantiates the third view with a shared byte trie and a sparse tree-Wasserstein loss.

2.3.4 Answer-Token Distillation

For multimodal question answering, distillation should focus on answer tokens, not prompt tokens [19].

In our implementation, the code computes KD only on supervised answer positions after masking ignored tokens, optionally removing EOS and truncating student and teacher answer spans to a common length.

In Chapter 3, these sets are denoted by Ω_b^{CE} for CE and $\Omega_{b,i}^{KD}$ for teacher-specific KD.

This gives a token-aligned distillation signal while remaining compatible with different tokenizers.

2.4 Multi-Teacher Distillation

2.4.1 Per-Teacher KD Loss

Let $\mathcal{T} = \{T_1, \dots, T_N\}$ be a pool of frozen teachers. For sample b , teacher T_i produces a per-teacher distillation loss $\mathcal{L}_{KD}^{(b,i)}$. In principle, one could average these losses uniformly,

$$\mathcal{L}_{KD}^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{KD}^{(b,i)}, \quad (2.8)$$

but this assumes every teacher is equally useful for every sample.

In practice, different teachers may provide conflicting guidance. Some teachers may help the supervised objective on a given sample, while others may push the student in a different direction [45]. This motivates a selective routing mechanism that decides, for each sample, which teachers should be active.

2.4.2 Learned Routing and Load Balancing

Recent sparse expert models route each sample or token through only a subset of available experts. This improves efficiency and allows specialized computation [43, 48, 49].

The same high-level idea is useful in multi-teacher distillation. Instead of averaging all teachers equally, a learned router can propose teacher weights from student-side features [40–42].

Let $\mathbf{h}^{(b)}$ denote a pooled student representation for sample b . A router network $G(\cdot)$ produces gate logits

$$\mathbf{r}^{(b)} = G(\mathbf{h}^{(b)}), \quad \tilde{\mathbf{r}}^{(b)} = \text{AdjustedScores}(\mathbf{r}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\tilde{\mathbf{r}}^{(b)}), \quad (2.9)$$

where $p_i^{(b)}$ is the initial routing weight for teacher T_i . In the current implementation, $\tilde{\mathbf{r}}^{(b)}$ includes expert bias, optional training noise and router-temperature scaling before softmax.

In sparse routing, these weights are typically refined by additional constraints:

1. **Top- k routing**, which keeps only the highest-scoring teachers.
2. **Capacity constraints**, which prevent one teacher from receiving too many assignments.
3. **Load balancing**, which discourages routing collapse onto a small subset of teachers.

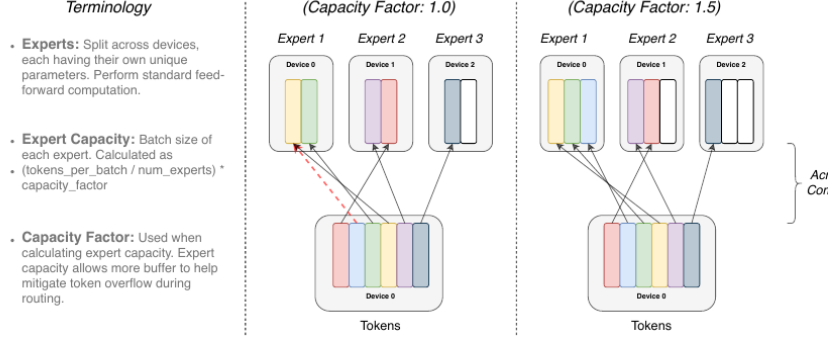


Figure 2.11: Illustration of sparse routing with expert capacity constraints, adapted from Switch Transformers [43]. Capacity factors control how many routed samples each expert can accept before overflow, which is directly analogous to the capacity-aware teacher routing used in GRACE.

Following the general MoE literature, one can summarize the balancing signal by a soft load $\bar{p}_i$ and a hard assignment load $\bar{q}_i$ after sparse routing, capacity control and fallback and define an auxiliary balancing objective of the form

$$\mathcal{L}_{bal} = N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (2.10)$$

This auxiliary term does not replace the KD objective; instead, it regularizes the router so that different teachers remain usable during training.

In GRACE, the code implements the router as a small student-conditioned MLP gate on top of pooled student hidden states. It then applies softmax weights, top- k sparsification and capacity filtering, with a short dense warmup before sparse routing starts. The full framework also supports optional balancing, entropy regularization and expert-bias updates.

When enabled, the expert bias term nudges the router toward under-used teachers in subsequent batches.

This makes the routing stage closer to sparse-expert training than to a static teacher average [43, 44, 48].

The next figure adds the specialization view. Sparse routing does not only save compute, it also lets different experts focus on different inputs.

Figures 2.11 and 2.12 summarize the sparse-routing ideas that motivate the teacher gate in GRACE.

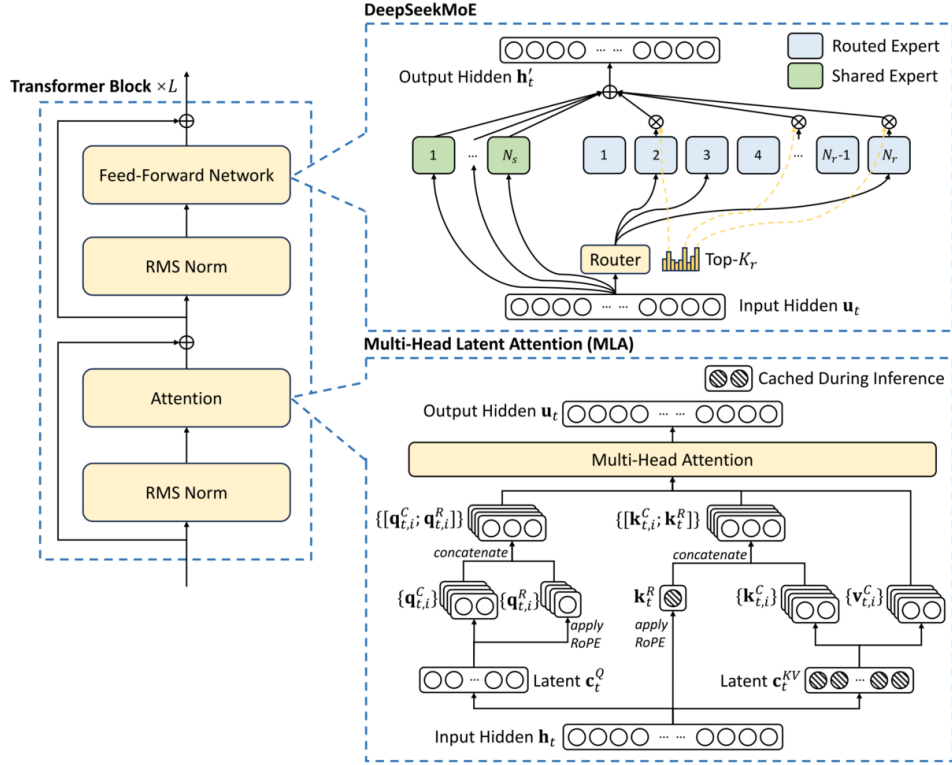


Figure 2.12: Example of sparse routing and expert specialization in a mixture-of-experts architecture [49, 50]. Although our setting is multi-teacher distillation, not token-level MoE language modeling, the same principles of learned routing, sparse activation and load balancing are directly relevant.

2.4.3 Gradient Alignment as a Routing Signal

Prior work uses gradient similarity in distillation as a signal for deciding whether teacher supervision helps or conflicts with the student objective [45]. If the teacher-induced update points in a similar direction to the supervised update, the teacher is more likely to be helpful. If the two directions conflict, teacher supervision may hurt learning on that sample [45].

In GRACE, this idea is applied only after routing. The router first decides which teachers remain available and Chapter 3 then refines those routed teachers with continuous gradient-based scores. The GRACE-specific details, including score smoothing, score softmax, router-score blending and uniform fallback, are therefore deferred to Chapter 3.

2.5 Related Work

2.5.1 Compact VLM Distillation and Compression

Knowledge distillation is a standard way to compress large models into smaller students while retaining part of the teacher’s predictive structure [11, 30]. In multimodal systems, prior work has used distillation to compress visual-linguistic models, transfer reasoning behavior, or improve compact vision-language students [20, 38, 39]. These studies show that small VLMs can benefit from teacher supervision, but most of them do not focus on heterogeneous teacher pools with mismatched tokenizers.

Compression by quantization is also important in practice, but it is a different intervention point: quantization changes the numerical precision of a trained model, while distillation changes how the student is trained [34–37]. This thesis therefore treats quantization as complementary deployment work and focuses the method contribution on distillation.

2.5.2 Cross-Tokenizer Distillation

Cross-tokenizer KD becomes necessary when teacher and student vocabularies do not match. ULD is a direct baseline in this setting because it compares sorted probability mass instead of token IDs [12]. More recent work also studies tokenizer-agnostic distillation through transport-based or alignment based formulations [15, 33, 47].

The main limitation of ranked-mass matching is that it discards structural relations between decoded token strings. Tree-structured transport retains that structure by comparing mass on shared branches of a hierarchy [14, 15]. This is the line of work most relevant

to our trie-Wasserstein loss: the student and teacher are not matched by token ID, but by mass placed on a shared UTF-8 byte trie.

2.5.3 Multi-Teacher Distillation and Adaptive Routing

Multi-teacher KD has been studied with fixed teacher averages, stochastic selection and meta-learned weighting rules [40–42]. These methods motivate the use of more than one teacher, but they do not solve the full combination of heterogeneous VLM teachers, tokenizer mismatch, and sparse routed weighting in one compact-VLM setting.

Sparse routing and load balancing are well established in mixture-of-experts models [43, 44, 48]. Our teacher gate borrows that routing view, but applies it to teacher selection, not to internal expert blocks. Gradient-based teacher filtering has also appeared in distillation through alignment-style objectives [45]. GRACE combines these ideas in a two-stage procedure: sparse student-conditioned routing first, then post-routing GRACE score refinement.

2.5.4 Relation to This Thesis

This thesis sits at the intersection of the three lines above. It studies compact generative VLM distillation, replaces ranked-mass ULD with a cross-tokenizer trie-Wasserstein loss in the main method and combines that loss with sparse multi-teacher routing plus gradient-based score refinement. Later chapters keep ULD, SHM-CKA and reinforced teacher selection as baselines, but the final method integrates the transport, routing, and alignment ideas into one framework.

Chapter 3

Proposed Method

3.1 Method Overview and Problem Formulation

We consider text-rich multimodal understanding tasks that require reading image text, understanding the question and generating an answer sequence [7–9]. The training set is

$$\mathcal{D} = \{(x_b, y_b)\}_{b=1}^B, \quad (3.1)$$

where x_b is a multimodal input and y_b is the target answer sequence.

Let S_θ denote the student VLM with trainable parameters θ and let

$$\mathcal{T} = \{T_1, T_2, \dots, T_N\} \quad (3.2)$$

be a pool of frozen teacher VLMs. For each sample b , the student produces logits $\mathbf{z}_s^{(b)}$ and teacher T_i produces teacher logits $\mathbf{z}_t^{(b,i)}$.

The goal of this thesis is not to pre-select a fixed teacher subset before training. Instead, we train the student with *Gradient-Routed Alignment with Capacity-aware Experts (GRACE)*, a routed multi-teacher distillation framework that refines sparse teacher assignments with gradient-based scores.

GRACE has five components:

1. A supervised student objective $\mathcal{L}_{CE}$.
2. A per-teacher trie-Wasserstein objective $\mathcal{L}_{KD}^{(b,i)}$.
3. A student-conditioned MLP gate.
4. Top- k routing, capacity control and router regularization.
5. GRACE score refinement based on pooled gradient cosine similarity.

At a high level, the gate first proposes dense teacher weights from pooled student hidden states. A short hard-routing warmup keeps all teachers available; after that, top- k and capacity constraints make the routing sparse. Each routed teacher then contributes a candidate KD loss and a GRACE score that measures directional agreement with the supervised CE update.

Before GRACE warmup, the method uses uniform KD weights over the teachers that remain after routing. After warmup, GRACE smooths the scores with an EMA, converts them to a score softmax, blends them with the routed weights, and falls back to a uniform distribution only when the score spread is too small to justify a strong preference.

The current repository objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD} + \lambda_{bal} \mathcal{L}_{bal} + \lambda_{ent} \mathcal{L}_{ent} + \lambda_z \mathcal{L}_z, \quad (3.3)$$

where λ_{bal} , λ_{ent} and λ_z are router regularization weights.

This design is closer in spirit to selective gradient-based distillation [45] and sparse expert routing [48]. Large-scale routed Transformer systems such as Switch Transformers show the same need for sparse assignment and overflow control [43].

Here we adapt that idea to the multi-teacher multimodal setting with token-level logit distillation.

The later sections define the trie-Wasserstein loss, routed teacher selection and GRACE score refinement in detail.

3.1.1 Overall Training Flow

Figure 3.1 gives a compact architecture view of the training step before the loss details below.

The training step follows four stages:

1. Run the student to obtain $(\mathbf{z}_s, \mathbf{H}_s, \mathcal{L}_{CE})$.
2. Compute gate logits $\mathbf{r}$, adjusted routing scores $\tilde{\mathbf{r}}$, soft routing weights $\mathbf{p}$ and route teachers under top- k and capacity constraints.
3. Compute routed per-teacher trie-Wasserstein losses $\mathcal{L}_{KD}^{(b,i)}$.
4. Compute pooled gradient-alignment scores, smooth them over time, blend them with the routed weights and apply fallback when needed.

Algorithm 1 summarizes one training step of GRACE before the later sections expand each component.

The final KD loss is formed sample-wise, $\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}$, then averaged over the batch. The full objective remains the sum of CE, routed KD and router regularizers.

Algorithm 1 One training step of GRACE

Require: mini-batch $\{(x_b, y_b)\}_{b=1}^B$, frozen teachers $\{T_i\}_{i=1}^N$

$(\mathbf{z}_s, \mathbf{H}_s, \mathcal{L}_{CE}) \leftarrow S_\theta(x)$

$\mathbf{h} \leftarrow \text{Pool}(\mathbf{H}_s); \mathbf{r} \leftarrow G(\mathbf{h})$

$\tilde{\mathbf{r}} \leftarrow (\mathbf{r} + \mathbf{b} + \boldsymbol{\xi})/\tau_g; \mathbf{p} \leftarrow \text{Softmax}(\tilde{\mathbf{r}})$

route teachers with current k_t (N during hard-routing warmup, else k), capacity and fallback to get $\hat{\mathbf{p}}$ and $\mathcal{R}_b$

for each routed teacher $i \in \mathcal{R}_b$ **do**

 compute teacher logits $\mathbf{z}_t^{(b,i)}$

 compute trie-Wasserstein loss $\mathcal{L}_{KD}^{(b,i)}$ on matched answer positions $\Omega_{b,i}^{KD}$

end for

if $t < \rho_g T$ **then**

$w^{(b)} \leftarrow \text{Uniform}(\mathcal{R}_b)$

else

 compute cosine scores $s_i^{(b)}$ from pooled CE and KD logit-gradient proxies

 update EMA-smoothed scores $\hat{s}_i^{(b)}$

$q^{(b)} \leftarrow \text{Softmax}(\beta \hat{\mathbf{s}}^{(b)})$ over $\mathcal{R}_b$

 blend $q^{(b)}$ with $\hat{\mathbf{p}}^{(b)}$ to get $w^{(b)}$

if $\max_{i \in \mathcal{R}_b} \hat{s}_i^{(b)} - \min_{i \in \mathcal{R}_b} \hat{s}_i^{(b)} < \epsilon$ **then**

$w^{(b)} \leftarrow \text{Uniform}(\mathcal{R}_b)$

end if

end if

$\bar{\mathcal{L}}_{KD}^{(b)} \leftarrow \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}$

$\mathcal{L} \leftarrow \mathcal{L}_{CE} + \alpha \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)} + \lambda_{bal} \mathcal{L}_{bal} + \lambda_{ent} \mathcal{L}_{ent} + \lambda_z \mathcal{L}_z$

update θ with $\nabla_\theta \mathcal{L}$

3.2 Trie-Wasserstein Cross-Tokenizer Distillation

3.2.1 Supervised Answer Tokens

We apply distillation only on supervised answer tokens, not on prompt tokens. For sample b , let Ω_b^{CE} denote the student positions whose labels are not ignored. For each teacher T_i , let $\Omega_{b,i}^{KD}$ denote the matched answer positions after:

1. Masking ignored labels on both student and teacher sides.
2. Optionally removing EOS from the student or teacher answer span.
3. Truncating both spans to a common length.

This produces one supervised student set for CE and one matched student-teacher set for each KD term.

3.2.2 Shared Byte Trie

For each sample b , token position $j \in \Omega_{b,i}^{KD}$ and teacher T_i , let $\mathbf{z}_{s,j}^{(b)} \in \mathbb{R}^{V_s}$ be the student logits and $\mathbf{z}_{t,j}^{(b,i)} \in \mathbb{R}^{V_i}$ be the teacher logits. The student and teacher probability distributions are

$$\mathbf{p}_{s,j}^{(b)} = \text{softmax}\left(\frac{\mathbf{z}_{s,j}^{(b)}}{\tau_s}\right), \quad \mathbf{p}_{t,j}^{(b,i)} = \text{softmax}\left(\frac{\mathbf{z}_{t,j}^{(b,i)}}{\tau_t}\right), \quad (3.4)$$

where τ_s and τ_t are the student and teacher temperatures.

The current implementation does not compare these distributions by token ID or sorted rank. Instead, the method decodes each single-token vocabulary entry to a string, converts that string to UTF-8 bytes, appends an EOS sentinel, and inserts the resulting byte path into one shared trie. If e is a trie edge at depth $d(e)$, its transport weight is

$$w(e) = \begin{cases} \rho^{d(e)-1}, & e \neq \text{TAIL}, \\ 1, & e = \text{TAIL}, \end{cases} \quad (3.5)$$

where $0 < \rho < 1$ is the edge-decay factor. Shared prefixes therefore receive the same early edges, while deeper trie disagreements are discounted. The synthetic TAIL edge is handled separately with weight 1 [14, 15].

Different tokenizers may stop at different depths, but they still share the same byte-prefix edges when their decoded token strings are related. The trie therefore **matches**

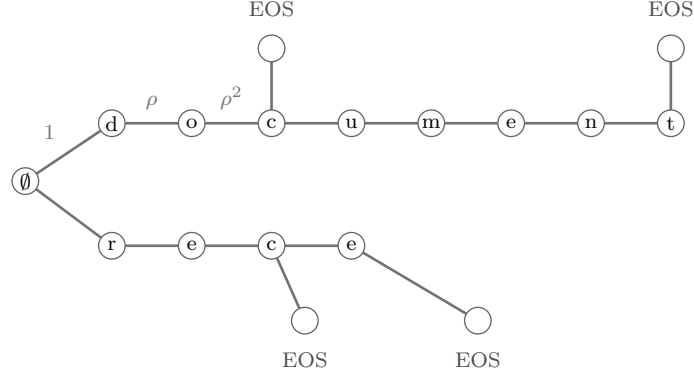


Figure 3.2: Shared-trie view of cross-tokenizer matching for `document/receipt` and `doc/rece`. Shared byte prefixes occupy common edges, while shorter teacher pieces terminate earlier at EOS.

probability mass on shared byte paths instead of token IDs. Figure 3.2 uses one simple example with student strings `document/receipt` and teacher strings `doc/rece`.

Figure 3.2 shows the main structural case: shorter teacher pieces terminate earlier with EOS, while longer student pieces continue deeper in the same trie, so the mismatch becomes structural instead of purely symbolic.

3.2.3 Sparse Top- k Support and Tail Edge

For efficiency, the loss keeps only the top- k_{TW} non-ignored student and teacher tokens at each supervised position. Let $\mathcal{K}_{s,j}^{(b)}$ and $\mathcal{K}_{t,j}^{(b,i)}$ be those retained token sets. The code then collapses the remaining probability mass into a dedicated TAIL edge:

$$r_{s,j}^{(b)} = 1 - \sum_{v \in \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v), \quad r_{t,j}^{(b,i)} = 1 - \sum_{v \in \mathcal{K}_{t,j}^{(b,i)}} p_{t,j}^{(b,i)}(v). \quad (3.6)$$

For any active trie edge e , the retained subtree masses are

$$m_{s,j}^{(b)}(e) = \sum_{v \in \mathcal{K}_{s,j}^{(b)}: e \in \text{path}(v)} p_{s,j}^{(b)}(v) + \mathbf{1}[e = \text{TAIL}] r_{s,j}^{(b)}, \quad (3.7)$$

$$m_{t,j}^{(b,i)}(e) = \sum_{v \in \mathcal{K}_{t,j}^{(b,i)}: e \in \text{path}(v)} p_{t,j}^{(b,i)}(v) + \mathbf{1}[e = \text{TAIL}] r_{t,j}^{(b,i)}. \quad (3.8)$$

The token-level trie-Wasserstein loss is then

$$\ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) = \sum_{e \in \mathcal{E}_{act,j}^{(b,i)}} w(e) \left| m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e) \right|, \quad (3.9)$$

where $\mathcal{E}_{act,j}^{(b,i)}$ contains the active student edges, active teacher edges and the TAIL edge.

Sparse trie-Wasserstein KD uses top- k support plus a TAIL edge

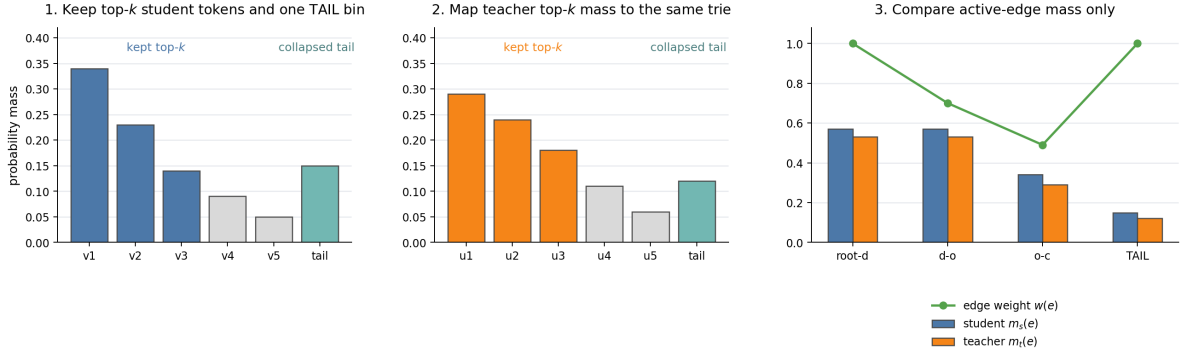


Figure 3.3: Sparse trie-Wasserstein support in the current implementation. The loss keeps only the top- k_{TW} student and teacher tokens at each supervised position, maps them to the shared trie and sends the remaining mass to one TAIL edge. This makes the computation depend on the active support, not the full vocabulary.

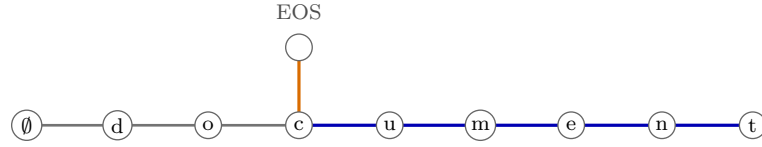


Figure 3.4: Trie split for `document` and `doc`. The shared prefix ends at `c`, where the teacher terminates at EOS and the student continues deeper in the trie.

The per-sample, per-teacher KD loss remains the mean over supervised answer tokens:

$$\mathcal{L}_{KD}^{(b,i)} = \frac{1}{|\Omega_{b,i}^{KD}|} \sum_{j \in \Omega_{b,i}^{KD}} \ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}). \quad (3.10)$$

Compared with the ULD baseline from Chapter 2, this loss compares mass on shared trie edges instead of ranked logits. It keeps the cross-tokenizer goal, but it uses a structured transport view that is closer to the actual tokenizer geometry, while the top- k +TAIL approximation keeps the cost bounded by the active support.

The sparse view explains why the loss remains practical for compact students: the code never builds a dense transport matrix and the TAIL edge captures the residual low-probability mass in one term.

Figure 3.4 connects the equations to one simple case. Here the shared prefix contributes little because both sides place mass on the same early edges. The main loss appears after `c`: the teacher sends mass to EOS, while the student sends mass to the continuation path, so the token-level term grows on the first mismatched edges.

Teacher / student top token	$\mathcal{L}_{ULD}$	ℓ_{TW}
doc / document	0.20	0.60
abc / def	0.20	3.29

Table 3.1: Illustrative comparison between ULD and trie-Wasserstein under the same ranked masses. ULD assigns the same loss to both rows, while trie-Wasserstein gives lower loss to the pair with a shared byte prefix.

Table 3.1 gives one small comparison against the ULD baseline. In both rows, the ranked masses are the same: the teacher places 0.8 on its top token and 0.2 on the residual tail, while the student places 0.9 on its top token and 0.1 on the residual tail. With $\rho = 0.5$ and $\text{top-}k_{TW} = 1$, ULD returns the same value in both cases because it only compares sorted mass. Trie-Wasserstein gives a much smaller loss to `doc` vs `document` than to `abc` vs `def` because the first pair shares a long byte prefix.

3.3 Learned Teacher Gate and Load Balancing

3.3.1 Student-Conditioned Router

When the teacher pool contains multiple models, the method first builds a student-conditioned router instead of averaging teachers uniformly.

Let $\mathbf{H}_s^{(b)}$ denote the student hidden states captured at the routing source layer for sample b . We pool these hidden states over supervised answer positions to obtain a routing feature

$$\mathbf{h}^{(b)} = \text{Pool}(\mathbf{H}_s^{(b)}) . \quad (3.11)$$

In code, this pool is the mean over supervised answer positions. If a sample has no supervised answer token, the gate falls back to the positions selected by the attention mask.

A student-conditioned MLP router $G(\cdot)$ then produces gate logits

$$\mathbf{r}^{(b)} = G(\mathbf{h}^{(b)}) . \quad (3.12)$$

The code then forms routing scores

$$\tilde{\mathbf{r}}^{(b)} = \frac{\mathbf{r}^{(b)} + \mathbf{b} + \boldsymbol{\xi}^{(b)}}{\tau_g}, \quad \mathbf{p}^{(b)} = \text{softmax}(\tilde{\mathbf{r}}^{(b)}) , \quad (3.13)$$

where $\mathbf{b}$ is the expert-bias vector, τ_g is the router temperature and $\boldsymbol{\xi}^{(b)}$ is Gaussian router noise during training. Here $p_i^{(b)}$ is the initial gate weight for teacher T_i .

3.3.2 Top-k, Capacity and Fallback

Let k_t denote the effective top- k value at training step t . The router is made sparse by applying top- k routing to the same routing scores $\tilde{r}_i^{(b)}$. Let $\mathcal{K}_b = \text{TopK}(\tilde{\mathbf{r}}^{(b)}, k_t)$ be the candidate teacher set for sample b . To prevent one teacher from receiving too many assignments, each teacher is additionally limited by a capacity

$$C_t = \left\lceil \phi \frac{Bk_t}{N} \right\rceil, \quad (3.14)$$

where B is the mini-batch size, N is the number of teachers, and $\phi > 0$ is the capacity factor.

Let $m_i^{(b)} \in \{0, 1\}$ denote the resulting assignment mask after top- k selection, capacity pruning and fallback. The constrained gate weights are

$$\hat{p}_i^{(b)} = \frac{m_i^{(b)} p_i^{(b)}}{\sum_{j=1}^N m_j^{(b)} p_j^{(b)}}. \quad (3.15)$$

If capacity pruning leaves a sample without any assigned teacher, the algorithm falls back to the teacher with the largest bias-adjusted routing score $\tilde{r}_i^{(b)}$.

This ensures that at least one teacher remains available before GRACE refinement.

The current implementation also uses a short hard-routing warmup. Let ρ_h be the hard-routing warmup ratio. During the first $\rho_h T$ steps, the router uses

$$k_t = N, \quad (3.16)$$

so all teachers remain available. After that point, the router switches to the configured sparse value $k_t = k$.

3.3.3 Balancing Loss and Expert Bias Update

The full GRACE framework regularizes the gate with a balancing loss that uses both soft routing weights and hard assignments.

Define the soft load

$$\bar{p}_i = \frac{1}{B} \sum_{b=1}^B p_i^{(b)} \quad (3.17)$$

and the hard assignment load

$$\bar{q}_i = \frac{\sum_{b=1}^B m_i^{(b)}}{\sum_{b=1}^B \sum_{j=1}^N m_j^{(b)}}, \quad (3.18)$$

The balancing objective is

$$\mathcal{L}_{bal} = N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (3.19)$$

The expert bias is updated from the per-batch assignment load ℓ_i to discourage teacher overuse:

$$b_i \leftarrow b_i - \eta_b \left(\ell_i - \frac{1}{N} \sum_{j=1}^N \ell_j \right), \quad (3.20)$$

where η_b is a small bias-update rate.

The current code also adds two small auxiliary router penalties:

- a negative-entropy term

$$\mathcal{L}_{ent} = -\frac{1}{B} \sum_{b=1}^B H(\mathbf{p}^{(b)}), \quad (3.21)$$

which discourages premature routing collapse.

- a router z -loss

$$\mathcal{L}_z = \frac{1}{B} \sum_{b=1}^B \left(\log \sum_{i=1}^N e^{r_i^{(b)}} \right)^2, \quad (3.22)$$

which keeps router logits numerically well behaved.

Figure 3.5 is a high-level sketch of the gate. The actual implementation applies LayerNorm $\rightarrow$ up projection $\rightarrow$ split into gate and value branches $\rightarrow$ SiLU(gate) $\odot$ value $\rightarrow$ down projection, then applies bias, top- k , capacity control, fallback and balance updates to the resulting teacher logits.

Figure 3.6 isolates the effect of the bias update. Figure 3.7 gives a routing example inspired by the routing-overflow and load-balancing behavior used in sparse expert models [43, 44]. The dense gate may prefer a small subset of teachers, but top- k selection, capacity constraints and fallback convert these scores into a sparse assignment pattern with explicit load statistics.

3.4 GRACE Score Refinement

3.4.1 Gradient Proxies

For each sample b , let $\mathcal{L}_{CE}^{(b)}$ be the student cross-entropy loss on supervised answer tokens. We define the pooled CE logit-gradient proxy as

$$\tilde{\mathbf{g}}_{CE}^{(b)} = \frac{1}{|\Omega_b^{CE}|} \sum_{j \in \Omega_b^{CE}} \nabla_{\mathbf{z}_{s,j}^{(b)}} \mathcal{L}_{CE}^{(b)}. \quad (3.23)$$

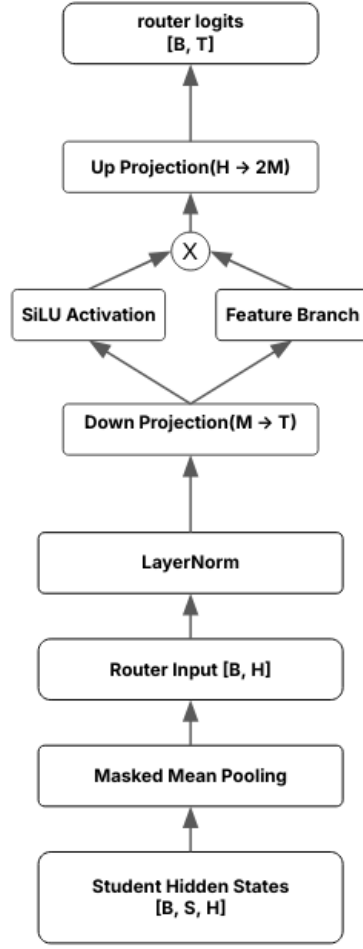


Figure 3.5: Conceptual sketch of the student-conditioned router in GRACE. The boxes summarize the routing components, while the exact execution order is given in the text below.

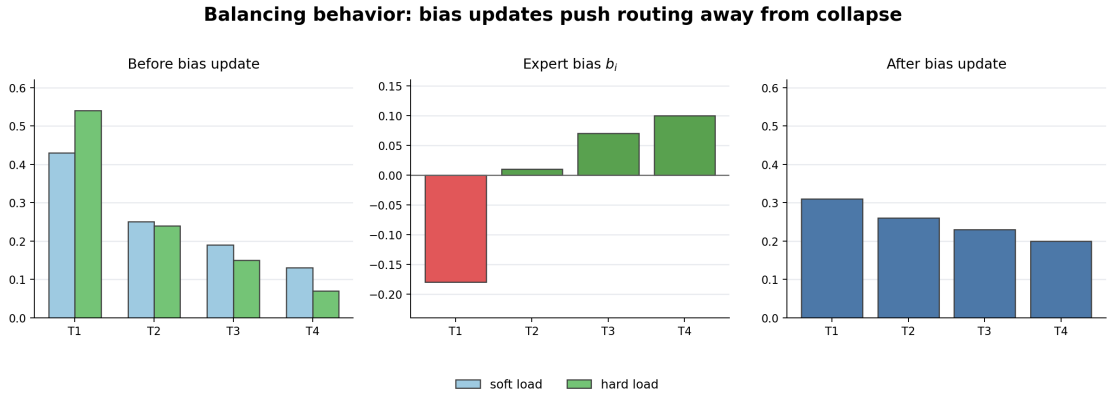


Figure 3.6: Bias-feedback behavior in GRACE under expert-bias updates. If one teacher receives too many assignments, its bias is reduced, while underused teachers receive higher bias. This shifts future routing toward a more balanced load.

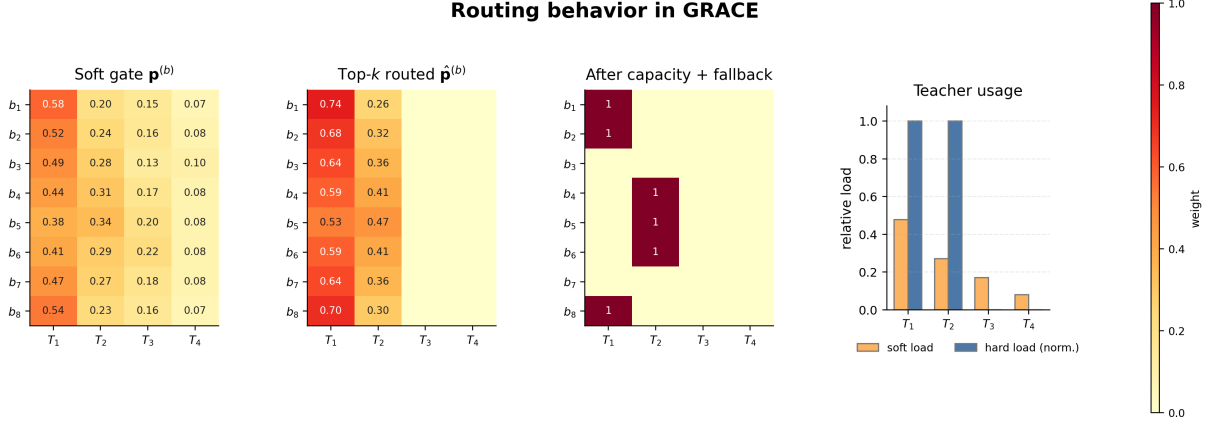


Figure 3.7: Routing dynamics in GRACE. Left to right: dense gate scores, top- k routed weights and the final assignment pattern after capacity control and fallback. The rightmost panel compares average soft load and normalized hard load, illustrating why balancing terms and expert-bias updates are needed.

In implementation, the code computes this gradient in closed form from the student probabilities and labels, which avoids a second backward pass through the full model.

For each teacher T_i , the pooled KD logit-gradient proxy is

$$\tilde{\mathbf{g}}_i^{(b)} = \frac{1}{|\Omega_b^{CE}|} \sum_{j \in \Omega_{b,i}^{KD}} \nabla_{\mathbf{z}_{s,j}^{(b)}} \mathcal{L}_{KD}^{(b,i)}. \quad (3.24)$$

The code computes this proxy directly from the selected distillation loss. In the default setting of this thesis, $\mathcal{L}_{KD}^{(b,i)}$ is the trie-Wasserstein loss in Equation 3.10. The KD sum uses the matched student-teacher positions $\Omega_{b,i}^{KD}$, but the normalization still uses the student CE count $|\Omega_b^{CE}|$.

3.4.2 GRACE Scores and Logged Active Flags

For each sample b and teacher T_i , we measure directional agreement between supervised learning and KD by cosine similarity:

$$s_i^{(b)} = \cos\left(\tilde{\mathbf{g}}_{CE}^{(b)}, \tilde{\mathbf{g}}_i^{(b)}\right). \quad (3.25)$$

The CE proxy averages over Ω_b^{CE} , while the KD proxy sums over $\Omega_{b,i}^{KD}$ and normalizes by $|\Omega_b^{CE}|$. This matches the current implementation.

The current implementation also records a thresholded activity flag

$$a_i^{(b)} = \mathbf{1}[s_i^{(b)} > \delta], \quad (3.26)$$

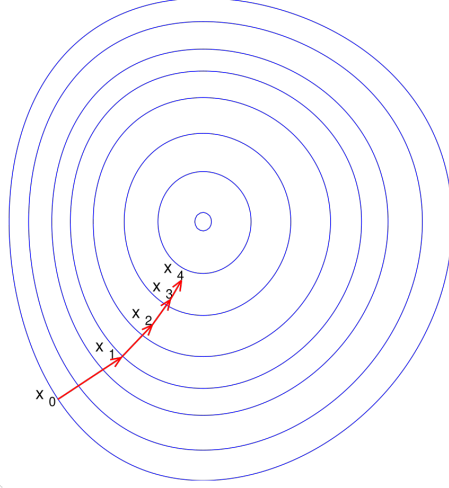


Figure 3.8: A directional view of gradient descent on a contour surface [51]. GRACE uses the same directional idea at the logit level: teachers whose KD update direction aligns better with the CE direction receive larger GRACE weights after warmup.

where δ is a logging threshold. These flags are useful for monitoring, but the GRACE weight update itself uses the continuous scores $s_i^{(b)}$.

Figure 3.8 gives the directional picture behind the cosine-based GRACE score.

3.4.3 Warmup, Score Smoothing and Gradient Weights

GRACE refinement starts only after a warmup phase. Let $\rho_g \in [0, 1]$ denote the GRACE warmup ratio with respect to the total number of training steps. During the first $\rho_g T$ steps, the current code uses a uniform KD distribution over the teachers that remain after routing:

$$u_i^{(b)} = \frac{\mathbf{1}[\hat{p}_i^{(b)} > 0]}{\sum_{j=1}^N \mathbf{1}[\hat{p}_j^{(b)} > 0]}. \quad (3.27)$$

After warmup, GRACE refines the routed teachers with a score-smoothed gradient distribution. Let $\bar{s}_i^{\text{prev}}$ be the running exponential moving average (EMA) of past batch means for teacher T_i and let γ be the EMA decay. The per-sample smoothed score is

$$\hat{s}_i^{(b)} = \gamma \bar{s}_i^{\text{prev}} + (1 - \gamma) s_i^{(b)}. \quad (3.28)$$

Figure 3.9 explains why GRACE uses EMA before the score softmax: **raw per-batch scores are noisy, while smoothed scores produce more stable teacher weights.**

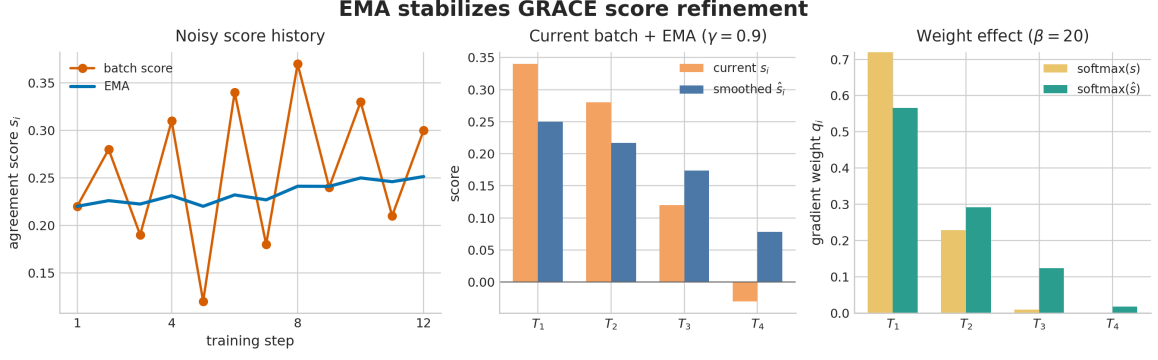


Figure 3.9: EMA smoothing in GRACE. The current batch score is blended with the running teacher score history before the GRACE softmax is applied. This damps batch-level noise and keeps the final gradient weights more stable.

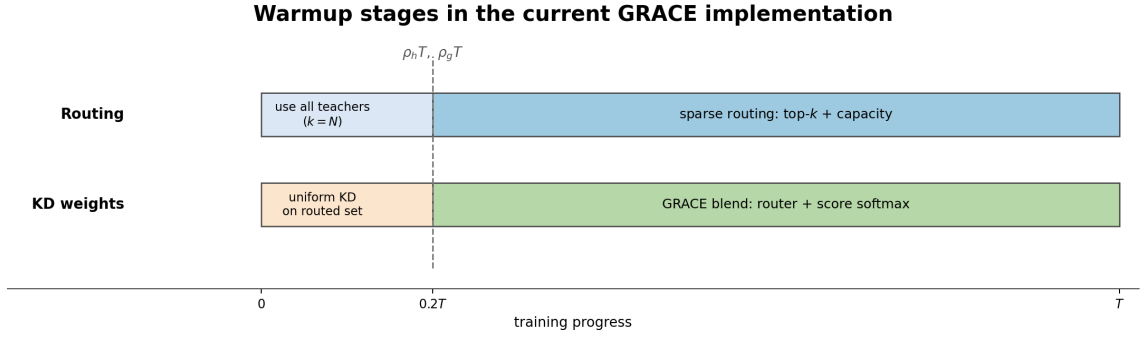


Figure 3.10: Warmup stages in the current implementation of GRACE. The hard-routing warmup starts with all teachers available, then switches to sparse top- k routing. GRACE warmup starts with uniform KD over the routed teachers, then activates score-based refinement after $\rho_g T$.

Let $\mathcal{R}_b = \{i \mid \hat{p}_i^{(b)} > 0\}$ be the routed teacher set. GRACE converts the smoothed scores into a gradient-weight vector

$$q_i^{(b)} = \frac{\exp(\beta \hat{s}_i^{(b)}) \mathbf{1}[i \in \mathcal{R}_b]}{\sum_{j=1}^N \exp(\beta \hat{s}_j^{(b)}) \mathbf{1}[j \in \mathcal{R}_b]}, \quad (3.29)$$

where β controls how sharply the highest-scoring teacher is preferred.

Figure 3.10 summarizes when hard routing and GRACE weighting become active in the current implementation.

3.4.4 Router-Gradient Blend and Uniform Fallback

After warmup, GRACE combines the routed weights and the gradient weights by a geometric blend. Let $\lambda \in [0, 1]$ be the router-blend exponent. The unnormalized blended weights are

$$\tilde{w}_i^{(b)} \propto \left(\hat{p}_i^{(b)}\right)^\lambda \left(q_i^{(b)}\right)^{1-\lambda}, \quad (3.30)$$

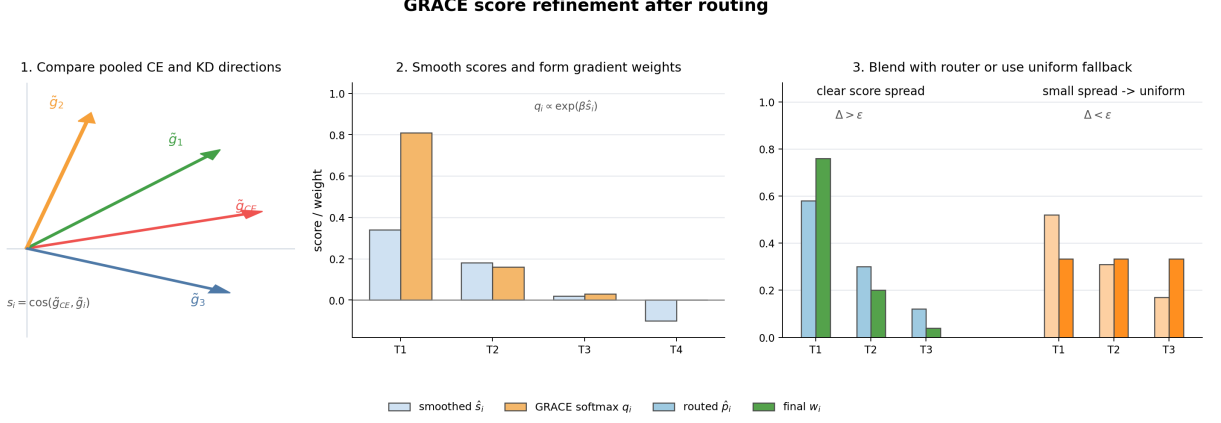


Figure 3.11: GRACE score refinement in GRACE. Left: pooled CE and KD gradients define per-teacher cosine scores. Middle: after warmup, the smoothed scores are converted into a GRACE softmax over the routed teachers. Right: the final KD weights blend the routed weights and GRACE weights, or become uniform when the score spread is below ϵ .

followed by normalization over the routed teachers:

$$w_i^{(b)} = \frac{\tilde{w}_i^{(b)} \mathbf{1}[i \in \mathcal{R}_b]}{\sum_{j=1}^N \tilde{w}_j^{(b)} \mathbf{1}[j \in \mathcal{R}_b]}. \quad (3.31)$$

The implementation uses a uniform fallback only when the smoothed scores are too close to justify a strong preference. Define the score spread

$$\Delta^{(b)} = \max_{i \in \mathcal{R}_b} \hat{s}_i^{(b)} - \min_{i \in \mathcal{R}_b} \hat{s}_i^{(b)}. \quad (3.32)$$

If $\Delta^{(b)} < \epsilon$, then GRACE falls back to

$$w_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b]}{|\mathcal{R}_b|}. \quad (3.33)$$

This keeps KD stable when the routed teachers have nearly indistinguishable GRACE scores.

Figure 3.11 presents the score update itself, while Figure 3.12 shows the resulting weighting patterns.

Figure 3.12 shows the same rule from a weight-update view. The routed gate weights are first proposed by the student-conditioned router, then refined by GRACE scores and finally converted into the effective KD weights used in Equation 3.34.

Figure 3.13 compresses the same update into one sample-wise path, from gate softmax to routed weights, GRACE softmax and final effective KD weights.

Figure 3.14 then shows the practical implication: the same teacher pool can lead to different final weights on different samples.

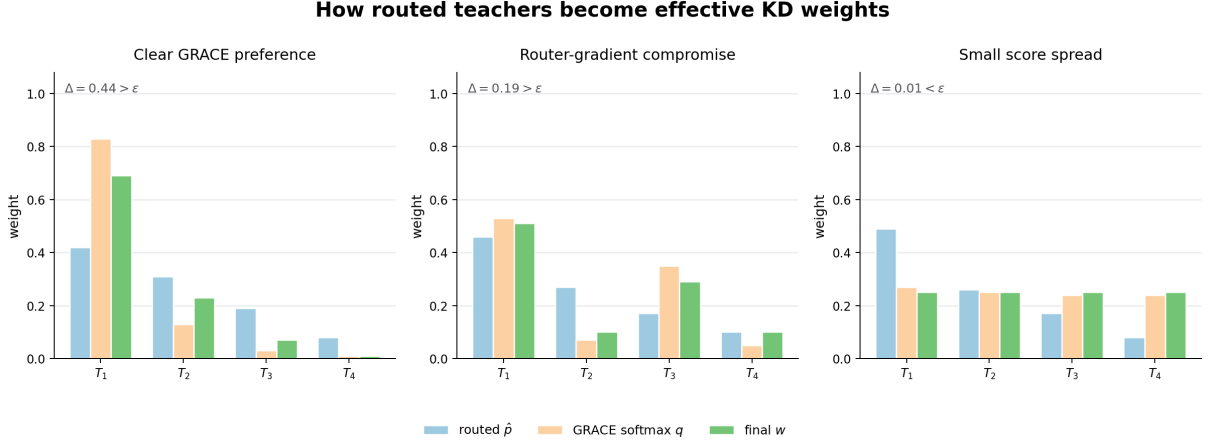


Figure 3.12: Three GRACE weighting cases in GRACE. When the GRACE scores show a clear preference, the final weights stay close to the strongest teacher. Under mixed router and GRACE preferences, the final weights become a compromise between both signals. When the score spread is very small, the method falls back to uniform weighting over the routed teachers.

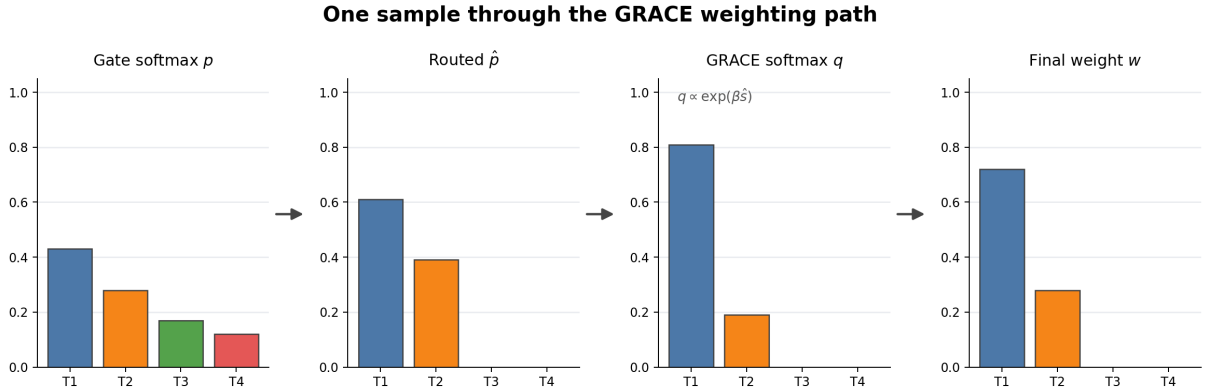


Figure 3.13: A sample-wise view of weight refinement in GRACE. The gate first proposes dense soft weights, routing constraints make them sparse, GRACE converts smoothed gradient scores into a second weight vector and the final KD weights blend both signals.

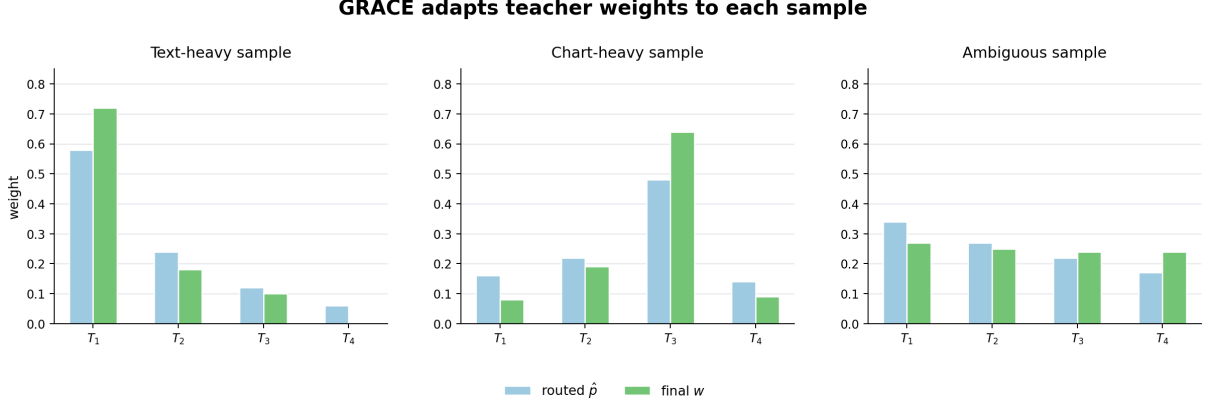


Figure 3.14: Sample-specific weighting in GRACE. The routed teacher set can remain similar across samples, while the final KD weights still shift according to the sample content and the GRACE score refinement stage.

3.4.5 Sample-Wise KD Aggregation

The routed KD loss for sample b is the weighted aggregation

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (3.34)$$

We then average these sample losses over the batch:

$$\mathcal{L}_{KD} = \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)}. \quad (3.35)$$

This rule has two stages: the gate defines the routed teacher set and GRACE refines those routed teachers into the final KD weights.

3.5 Final Objective

The final optimization objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD} + \lambda_{bal} \mathcal{L}_{bal} + \lambda_{ent} \mathcal{L}_{ent} + \lambda_z \mathcal{L}_z, \quad (3.36)$$

where $\mathcal{L}_{CE}$ is the standard supervised next-token cross-entropy loss and $\mathcal{L}_{KD}$ is defined in Equation 3.35. The balance term $\mathcal{L}_{bal}$ is defined in Equation 3.19; the entropy term $\mathcal{L}_{ent}$ and router z -loss $\mathcal{L}_z$ are small penalties on the teacher gate.

Equation 3.36 keeps the student anchored to ground-truth supervision while adding routed KD and small router regularizers. Compared with static multi-teacher averaging, this objective makes the teacher contribution adaptive at the sample level. Compared with

fixed teacher-subset selection, it avoids committing to one subset for the whole run [40, 41, 45].

Chapter 4

Evaluation Methodology

4.1 Experimental Setup

All experiments follow a shared evaluation protocol for fair comparison across methods.

The study focuses on two compact student models, **SmolVLM-500M** and **SmolVLM-256M**, evaluated on text-rich multimodal benchmarks [25].

We evaluate all models with **VLM-Eval-Kit** [52]. In brief, the toolkit works as follows:

1. Load a benchmark split and apply its prompt template or input format.
2. Run model inference through a unified model wrapper interface.
3. Post-process raw generations into normalized answers.
4. Compute dataset-specific scores (e.g., VQA Accuracy, ANLS, Relaxed Accuracy) and aggregate final results.

This unified pipeline ensures a consistent evaluation procedure across all compared models.

4.2 Datasets

We evaluate three text-centric VQA benchmarks: **TextVQA**, **DocVQA** and **ChartQA**.

The goal is to measure whether the model can **read textual content in images, understand questions and perform basic reasoning** across different visual domains [7–9].

We chose these datasets because they require optical character recognition (OCR)-heavy understanding, cover complementary visual domains and include retrieval, layout understanding and numerical reasoning [7–9].

4.2.1 Dataset Descriptions

TextVQA [7]: evaluates question answering on natural images containing scene text. It focuses on reading text in-the-wild and combining it with question understanding. *Sample*



Figure 4.1: Example image for a TextVQA-style question [53]. The task depends on reading scene text in a natural image and grounding the answer in that text.

question: “What word is written on the stop sign?” *Sample answer*: “No word is visible on this sign.”

DocVQA [8]: evaluates VQA on real document images (scans/photos), where layout, paragraphs, tables and forms are important. It emphasizes document structure understanding in addition to OCR. *Sample question*: “What is the invoice number shown in this document?” *Sample answer*: “4572.”

ChartQA [9]: evaluates question answering over charts. It tests chart reading plus numerical and logical reasoning to derive the final answer. *Sample question*: “Which year has the highest sales value in the chart?” *Sample answer*: “The second (middle) bar has the highest value.”

Figures 4.1, 4.2, and 4.3 show the three visual domains covered by the evaluation.

4.2.2 Dataset Size and Splits

Table 4.1 lists the split sizes used throughout the evaluation chapter.

4.3 Training Configuration

Unless otherwise stated, the main distillation experiments use a shared training configuration so that differences in performance can be attributed to the distillation strategy, not unrelated optimization choices.



Figure 4.2: Example document image for a DocVQA-style question [54]. Unlike TextVQA, the answer depends on document layout, local text spans and structured fields.

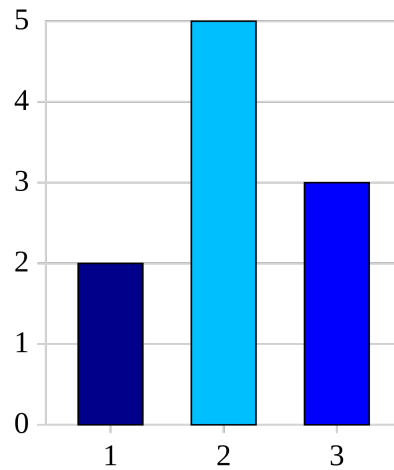


Figure 4.3: Example chart image for a ChartQA-style question [55]. This benchmark adds visual reading and numerical comparison on top of text extraction.

Dataset	Train	Val	Test	Total
TextVQA	34,602	5,000	5,734	45,336
DocVQA	39,463	5,349	5,188	50,000
ChartQA-H (human)	7,398	960	1,250	9,608
ChartQA-M (machine)	20,901	960	1,250	23,111
ChartQA (H+M, combined)	28,299	1,920	2,500	32,719

Table 4.1: Train, validation and test split sizes for TextVQA, DocVQA and ChartQA [7–9]. ChartQA is also broken down into its human and machine-generated subsets.

We run all training jobs on a single high-memory **NVIDIA H200** graphics processing unit (GPU) on **Modal** or on comparable single-GPU **Vast.ai** instances, depending on availability. A typical full H200 run takes about 7 hours and costs about 21 USD on Vast.ai, excluding the separate cost of offline teacher-logit generation and remote storage. Distillation training uses **DeepSpeed ZeRO Stage 2**, **Brain Floating Point 16 (bf16)** mixed precision, **TensorFloat-32 (TF32)** matrix multiplication and gradient checkpointing to control memory use while keeping the student fully trainable and all teachers frozen [56, 57].

Before training, we convert all benchmark samples into a unified **LLaVA-style multimodal instruction format** [19]. Each example is stored as an image plus a short user-assistant dialogue, so TextVQA, DocVQA and ChartQA can share the same data loader, prompt template and answer supervision interface.

Listing 4.1 shows the structure used in our preprocessing pipeline.

```

1 {
2   "image": "sample_000123.jpg",
3   "conversations": [
4     {"from": "human", "value": "<image>\nWhat is the invoice number?"},
5     {"from": "gpt", "value": "4572"}
6   ]
7 }
```

Listing 4.1: Representative LLaVA-style sample used after dataset conversion.

4.3.1 Offline Teacher-Logit Preparation

Teacher logits are generated *offline* before distillation. An offline caching script runs each frozen teacher on every training example, keeps only the supervised answer-token positions and writes one `.pt` file per sample under a teacher-specific directory. Each record stores

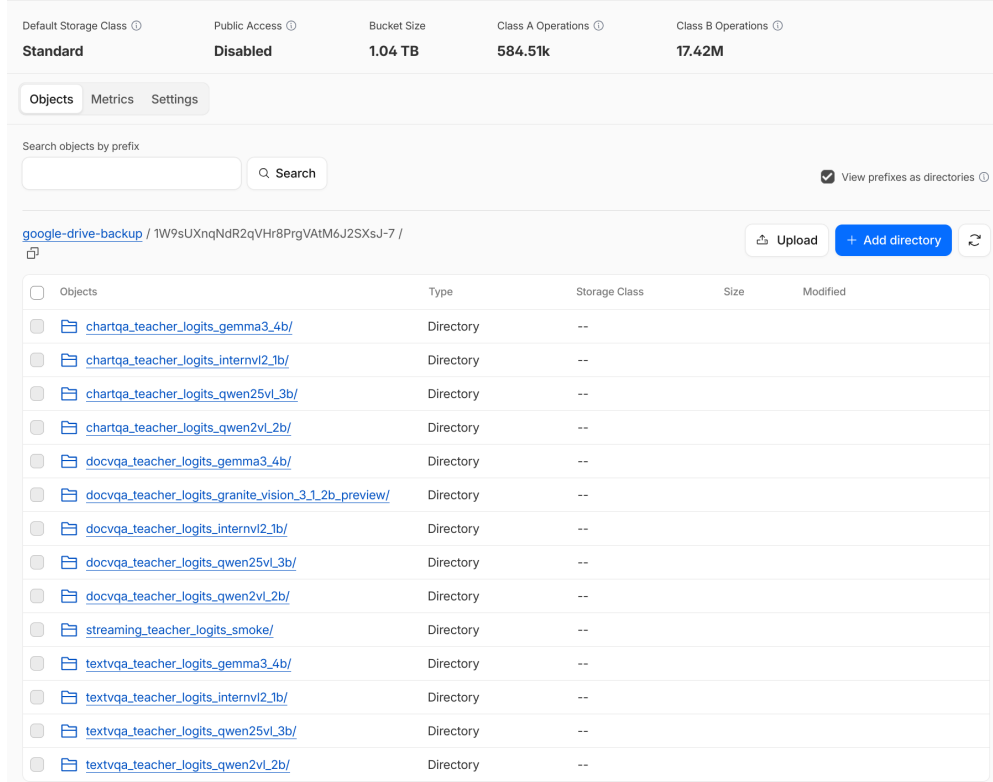


Figure 4.4: Cloudflare R2 bucket used for offline teacher-logit storage. The screenshot shows the large bucket footprint and the dataset-specific teacher-logit prefixes, which motivates remote streaming instead of copying the full cache onto each training instance.

the sample index, the selected teacher logits and the corresponding supervised labels. This keeps the online training loop simple: the student only reads precomputed teacher targets instead of running the teachers again.

After generation, we upload the teacher-logit directories to a **Cloudflare R2** bucket using its S3-compatible object-storage interface [58, 59]. In our setup, the bucket already exceeds **1 TB**, so copying the whole cache to a rented GPU instance is impractical. This is especially relevant on services such as Vast.ai, where local NVMe space is usually much smaller than the full teacher logit store.

Figure 4.4 shows the remote bucket layout used to store the offline teacher caches.

4.3.2 Remote Logit Streaming During Training

The current main training path does *not* predownload the full cache. When the launcher receives an R2 root `distill_multi_teachers.sh`, the trainer switches to remote lazy

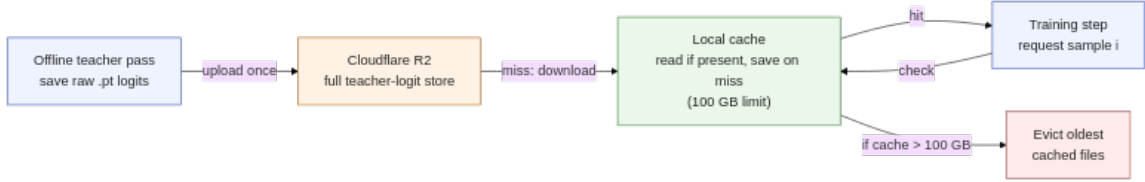


Figure 4.5: Training with remote teacher logits. Offline caches are uploaded to R2, fetched lazily during training and kept in a bounded local cache with oldest-file eviction. This setup avoids copying a multi-terabyte teacher store onto each rented GPU instance.

loading. During training, `StreamingTeacherLogitsCache` resolves the teacher-specific remote path

`remote_root/<dataset>_teacher_logits_<alias>/<teacher-slug>/<index>.pt,`

then fetches files lazily from R2 through `s3fs` and `WholeFileCacheFileSystem` [60, 61]. The repository also includes a repacking utility for MDS, but the current main training path reads the raw per-sample `.pt` cache directly.

To keep disk usage bounded, the loader uses a **100 GB** local cache. The loader caches each file only after first access. After a file is opened, the loader updates its access time. When the total cache size exceeds the limit, the loader removes the oldest cached files first. In effect, the local cache behaves as a small least-recently-used style buffer over the much larger R2 store. A per-teacher file lock also guards cache metadata updates, so concurrent dataloader workers do not corrupt the cache state.

Figure 4.5 summarizes the remote-cache training path used in the current implementation.

Table 4.2: Key hyperparameters, systems settings and cache settings for the current three-teacher DocVQA distillation script. The values come from the active launch path documented in this chapter, while Chapter 5 additionally reports broader completed runs.

Setting	Value
Student model	HuggingFaceTB/SmolVLM-500M-Instruct
	Qwen/Qwen2.5-VL-3B-Instruct
Teacher models	Qwen/Qwen2-VL-2B-Instruct
	ibm-granite/granite-vision-3.1-2b-preview
Teacher weighting strategy	routing (GRACE path)
Distillation loss	Trie-Wasserstein
Trie-Wasserstein rho	0.7
Trie-Wasserstein top- k	64

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Table 4.2 continued from the previous page.

Setting	Value
Training dataset	DocVQA
Validation split	validation
Hardware and runtime	Single-GPU runs on Modal (H200) or Vast.ai
Typical run cost	About 7 hours on H200, about 21 USD on Vast.ai
Distributed training	DeepSpeed ZeRO-2
Optimizer	adamw_torch
Adam betas	(0.9, 0.98)
Adam epsilon	1×10^{-7}
Weight decay	0.0
Numerical precision	bf16 enabled, half precision (fp16) disabled, TF32 enabled
FlashAttention-2	Enabled
Training data format	LLaVA-style multimodal instruction JSON
Teacher-logit storage	Cloudflare R2 object storage
Teacher-logit format	Raw per-sample .pt files
Local cache directory	/tmp/teacher-logits-streaming for R2 reads /workspace/cache for local-cache reads
Local cache limit	100 GB
Max sequence length	16384
Training epochs	2
Per-device batch size	80
Gradient accumulation steps	1
Effective batch size	80
Base learning rate	1×10^{-5}
Vision learning rate	2×10^{-6}
Connector learning rate	1×10^{-5}
Learning-rate schedule	Cosine decay with warmup ratio 0.03
Gradient checkpointing	Enabled
Frozen modules	None
Dataloader workers	4
Logging and checkpointing	Log every 5 steps; save every 150 steps
Student/teacher temperature	$\tau_s = \tau_t = 1.0$
KD weight	$\alpha = 0.5$
Top- k routing	$k = 2$
Capacity factor	$\phi = 1.25$
Teacher-gate temperature	1.5
Teacher-gate noise std	0.01
Balance weight	$\lambda_{bal} = 0.01$

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Table 4.2 continued from the previous page.

Setting	Value
Gate entropy weight	$\lambda_{ent} = 10^{-3}$
Router z -loss weight	$\lambda_z = 10^{-3}$
Bias-update rate	$\eta_b = 5 \times 10^{-4}$
Hard-routing warmup ratio	$\rho_h = 0.2$
GRACE warmup ratio	$\rho_g = 0.01$
GRACE softmax beta	$\beta = 10$
GRACE blend exponent	$\lambda = 0.4$
Score-spread fallback threshold	$\epsilon = 0.001$
Alignment threshold	$\delta = -0.005$
Seed	42
Image resolution / visual token budget	Processor defaults; no manual override

4.4 Evaluation Metrics

We report benchmark-specific official metrics for each dataset:

- **TextVQA: VQA Accuracy.** For a predicted answer $\hat{a}$, the per-question score is

$$s_{\text{VQA}} = \min \left(\frac{\# \text{annotators that provided } \hat{a}}{3}, 1 \right),$$

and the final score is the mean over all questions [7].

- **DocVQA: ANLS** (Average Normalized Levenshtein Similarity):

$$\text{NLS}(a, \hat{a}) = 1 - \frac{d_{\text{Lev}}(a, \hat{a})}{\max(|a|, |\hat{a}|)}, \quad s_{\text{ANLS}} = \begin{cases} \text{NLS}(a, \hat{a}), & \text{NLS} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

The final ANLS is the average score over all samples [8].

- **ChartQA: Relaxed Accuracy.** A prediction is correct if it matches exactly for text answers, or if the relative error is within 5% for numeric answers:

$$\frac{|\hat{y} - y|}{|y|} \leq 0.05.$$

The final score is the proportion of correct predictions [9].

Group	Methods	Teacher setup	Purpose
Non-distillation references	Original, full finetuning	No teacher	Student-only reference points
Same-family baselines	Forward KL, Reverse KL, Jensen-Shannon	SmolVLM2-2.2B-Instruct	Standard single-teacher objectives within the SmolVLM family
Different-family single-teacher baselines	ULD	Qwen2-VL-2B-Instruct	Cross-family transfer without multi-teacher routing
Heterogeneous multi-teacher baselines	Uniform averaging, SHM-CKA, reinforced selection, GRACE	Mixed teacher pool from multiple VLM families	Static averaging vs. hidden-state matching vs. adaptive teacher assignment

Table 4.3: Baseline groups used in the evaluation chapter. The grouping separates student-only references, same-family KD, cross-tokenizer single-teacher transfer and heterogeneous multi-teacher methods.

4.5 Baselines and Compared Methods

For consistency with Chapter 5, we organize the compared methods into four groups: non-distillation references, same-family distillation baselines, different-family single-teacher baselines, and heterogeneous multi-teacher baselines.

Table 4.3 summarizes the role of each baseline family before the method-specific descriptions below.

- **Non-distillation references.** The original student and the full-finetuning baseline provide the student-only reference points for each benchmark.
- **Same-family baselines.** Forward Kullback-Leibler (KL), reverse KL and Jensen-Shannon divergence all use **SmolVLM2-2.2B-Instruct** as a same-family teacher [11, 25, 46, 62, 63]. They isolate standard teacher-student transfer when tokenizer, vocabulary and output space are already matched.
- **Different-family single-teacher baselines.** This group uses one teacher outside the SmolVLM family. ULD is the main cross-tokenizer logits baseline [12].
- **Heterogeneous multi-teacher baselines.** This group uses teacher pools drawn from different VLM families. Uniform averaging keeps fixed equal weights. SHM-CKA adds hidden-state matching over the same pool. Reinforced selection learns a stochastic teacher policy with REINFORCE [40, 64]. GRACE adds sparse routing, trie-Wasserstein KD and alignment-aware reweighting [15, 33, 41, 42].

For the trainable competitor methods, we keep the same supervised term $\mathcal{L}_{CE}$. The logits-only baselines differ only in the KD signal, but SHM-CKA also adds a hidden-state alignment term and reinforced selection also adds a REINFORCE-style policy term [40, 64, 65]. In the descriptions below, γ denotes a generic KD mixing coefficient; α remains reserved for the current repository training script in Table 4.2.

The compared methods are:

- **Original:** direct evaluation of the released student, with no additional optimization. This measures the native capability of the compact model before any task adaptation or distillation.
- **Full finetuning:** uses $\mathcal{L} = \mathcal{L}_{CE}$ with no teacher, so it isolates the benefit of external supervision.
- **Forward KL:** the standard same-family KD baseline, with $\mathcal{L}_{KD} = \text{KL}(p_t \| p_s)$ [11]. It tests soft-target transfer when tokenizer and vocabulary are already matched.
- **Reverse KL:** uses $\mathcal{L}_{KD} = \text{KL}(p_s \| p_t)$ [63, 66]. This variant is more mode-seeking and serves as a deliberately harder divergence baseline.
- **Jensen-Shannon:** uses the symmetric divergence $\frac{1}{2}\text{KL}(p_s \| m) + \frac{1}{2}\text{KL}(p_t \| m)$, where $m = \frac{1}{2}(p_s + p_t)$ [46, 63]. It tests whether a bounded symmetric objective is more stable than KL.
- **ULD:** uses $\mathcal{L}_{KD} = \|\pi(p_s) - \pi(p_t)\|_1$ [12]. This is the main cross-tokenizer logits baseline because it compares ranked probability mass, not token IDs.
- **SHM-CKA:** keeps the same heterogeneous teacher pool as GRACE, but replaces routed weighting with a per-teacher ULD logits loss plus a hidden-state CKA penalty [12, 33, 65]. It tests whether direct representation alignment is sufficient without routed GRACE weighting.
- **Uniform multi-teacher:** averages the heterogeneous teacher losses with fixed equal weights, $\mathcal{L}_{KD} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{KD}^{(i)}$ [40, 41]. It isolates the effect of teacher diversity without adaptive assignment.
- **Reinforced selection:** replaces deterministic routing with a stochastic policy that samples teachers and updates the selector with REINFORCE [40, 64]. This baseline is close in spirit to adaptive selection, but it does not use GRACE score refinement.

For SHM-CKA, the historical training code first loads an offline CKA matrix and, for each student layer s , selects the top- k teacher layers $\mathcal{T}_s$. Their CKA scores are normalized into soft weights $\alpha_{s,t}$. The per-teacher layer loss is

$$\mathcal{L}_{\text{layer}}^{(i)} = \frac{1}{|S|} \sum_{s \in S} \sum_{t \in \mathcal{T}_s} \alpha_{s,t} (1 - \sqrt{\text{CKA}(\mathbf{H}_s, \mathbf{H}_t)}). \quad (4.1)$$

Each teacher also contributes its own ULD logits loss $\mathcal{L}_{\text{ULD}}^{(i)}$. In the historical single-teacher code, the default fixed-weight objective gives equal weight to CE, logits KD and layer matching:

$$\mathcal{L}_{\text{fixed}} = \frac{1}{3} (\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{ULD}} + \mathcal{L}_{\text{layer}}). \quad (4.2)$$

That code also supports an alternative adaptive weighting scheme, but the fixed equal-mean form above is the default historical setting. The original soft top- k matching path was implemented for single-teacher runs, so the heterogeneous baseline in this thesis keeps the same hidden-state idea per teacher, averages the logits and layer terms across the teacher pool,

$$\bar{\mathcal{L}}_{\text{ULD}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{ULD}}^{(i)}, \quad \bar{\mathcal{L}}_{\text{layer}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{layer}}^{(i)}, \quad (4.3)$$

then applies the same equal-mean combination

$$\mathcal{L}_{\text{SHM-CKA}} = \frac{1}{3} (\mathcal{L}_{\text{CE}} + \bar{\mathcal{L}}_{\text{ULD}} + \bar{\mathcal{L}}_{\text{layer}}). \quad (4.4)$$

For reinforced selection, the policy uses the same pooled student hidden state as a sample descriptor, then concatenates per-teacher summary features:

$$\mathbf{z} = [\text{Pool}(\mathbf{H}_s); \mathbf{f}_1; \dots; \mathbf{f}_M], \quad \mathbf{f}_i = [\mathcal{L}_{\text{KD}}^{(i)}, c_i, e_i], \quad (4.5)$$

where c_i is the teacher’s mean token confidence and e_i is its normalized token entropy over supervised positions. A linear policy then predicts independent Bernoulli probabilities

$$\boldsymbol{\pi} = \sigma(W \text{LN}(\mathbf{z}) + b). \quad (4.6)$$

During the warmup stage, all teachers stay active. After warmup, the baseline samples $m_i \sim \text{Bernoulli}(\pi_i)$, falls back to the highest-probability teacher if $\sum_i m_i = 0$ and normalizes the selected mask into weights

$$w_i = \frac{m_i}{\sum_{j=1}^M m_j}. \quad (4.7)$$

The multi-teacher KD term is the weighted average

$$\mathcal{L}_{\text{KD}} = \sum_{i=1}^M w_i \mathcal{L}_{\text{KD}}^{(i)}. \quad (4.8)$$

In the current launch script, $\mathcal{L}_{KD}^{(i)}$ is the per-teacher trie-Wasserstein loss. The selector is trained with a REINFORCE-style term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (4.9)$$

where b is an EMA reward baseline. The current script uses $r = -\mathcal{L}_{CE} - \frac{1}{\sum_i m_i} \sum_i m_i \mathcal{L}_{KD}^{(i)}$, so the total reinforced-selection objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (4.10)$$

Unless noted otherwise, all baselines reuse the same data conversion pipeline, hardware, optimizer family and checkpointing protocol described earlier in this chapter. Methods that do not use routing simply omit the GRACE-specific terms and retain the corresponding single-teacher or uniform multi-teacher objective from the cited source.

Chapter 5

Experiment Results

5.1 Main Comparison Results

This section compares the proposed method with the baselines on TextVQA, DocVQA and ChartQA.

5.1.1 Baseline Results

Before applying adaptation or distillation, we report the original performance of the two student models, **SmolVLM-500M** and **SmolVLM-256M**.

These baseline results establish the starting point for later comparison.

We obtain all results with **VLM-Eval-Kit** [52] under the same evaluation pipeline described in Chapter 4.

Table 5.1 gives the student-only starting point.

The teacher pool is substantially stronger than the compact students, which is what makes distillation meaningful in the first place.

Table 5.2 provides the corresponding teacher-side reference.

5.1.2 Overall Comparison

Tables 5.3, 5.4 and 5.5 summarize the main comparison results across original models, direct fine-tuning, standard logits distillation baselines and GRACE.

Unless stated otherwise, the main comparison tables report the strongest completed run for each method and benchmark. In particular, the adaptive heterogeneous multi-teacher rows, namely SHM-CKA, reinforced selection and GRACE, use the four-teacher pool {Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, InternVL3-1B, Gemma-3-4B-it}. The Uniform baseline is reported from a cheaper two-teacher run with {Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct}. Chapter 4 documents the current three-teacher DocVQA launch script for systems and reproducibility details.

We report the baselines in four groups.

Model	TextVQA	DocVQA	ChartQA
SmolVLM-500M	56.75	59.52	62.80
SmolVLM-256M	47.17	48.64	45.04

Table 5.1: Original benchmark results of the two SmolVLM students on TextVQA, DocVQA and ChartQA. These scores are the student-only reference point for all later fine-tuning and distillation comparisons.

Teacher Model	TextVQA	DocVQA	ChartQA
Qwen2.5-VL-3B-Instruct	79.30	93.90	94.24
InternVL3-1B	74.10	81.90	75.30
Gemma-3-4B-it	57.80	75.80	63.60
Qwen2-VL-2B-Instruct	79.70	90.10	89.28
Granite Vision 3.1 2B	88.00	76.00	86.00

Table 5.2: Original benchmark results of the teacher and large reference models used in this thesis. The table establishes the teacher-side reference range used later in the gap-to-teacher analysis.

- The **non-distillation references** are the original student and the full-finetuning baseline.
- The **same-family baselines** use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon divergence [11, 30, 33].
- The **different-family single-teacher baselines** use a teacher outside the SmolVLM family. This group includes the ULD baseline [12].
- The **heterogeneous multi-teacher baselines** use teacher pools drawn from different VLM families, including uniform averaging, SHM-CKA, reinforced selection and GRACE [40–42, 64].

We report each configuration as a single run. Multi-teacher training is expensive in both GPU time and teacher-logit storage, so standard deviations are not available for every method.

Because each multi-teacher run requires large offline logit caches and long single-GPU training, we report one fully trained run per method. A typical H200 run takes about 7

hours, which is about 21 USD on Vast.ai before storage and offline-logit generation cost. We therefore treat very small differences cautiously, focus on consistent multi-point gaps across datasets and student sizes and leave broader sweeps for later work.

DocVQA shows the same broad pattern, but the gains from adaptive weighting are even larger for the 500M student.

ChartQA is the hardest benchmark in our study. It is also the clearest case where stronger multi-teacher supervision matters.

The tables above show the end-point accuracy. The optimization path is also different: GRACE is visibly more stable than static uniform averaging.

Several trends are consistent across the three benchmarks:

- same-family logits baselines remain weak. Forward KL gives only small gains, reverse KL is consistently poor and Jensen-Shannon is unstable. Simply matching a same-family teacher distribution is not enough for these compact students.
- **cross-family ULD is already stronger than the same-family logits losses on all three tasks.** This shows that cross-tokenizer distillation is more useful here than vocabulary-matched KL. Once multiple heterogeneous teachers are added, uniform averaging gives another clear jump, which means teacher diversity itself is already valuable.
- **the remaining jump from the single-teacher ULD baseline to GRACE comes from the full recipe: a trie-Wasserstein KD loss, sample-dependent sparse routing and GRACE score refinement after routing.**
- **GRACE consistently improves on the uniform multi-teacher baseline for both student sizes.** The gain is moderate on TextVQA and DocVQA, but larger on ChartQA. This suggests that adaptive weighting helps most when the task mixes visual reading, numerical reasoning and teacher specialization.

Figure 5.1 gives the same comparison in a more compact visual form. The red bars stay above the blue and yellow bars on every task, and the separation is largest on ChartQA for both students.

Figure 5.2 complements the final benchmark tables with one illustrative optimization trace. **A representative GRACE run also follows a more stable training trajectory.**

Figure 5.3 shows the contrast directly. The figure is best read as a diagnostic contrast. In one routed run without the full GRACE recipe, the router drifts toward one teacher. In a

Category	Method	Teacher	Student	TextVQA		
Non-distillation references						
Original	None	None	SmolVLM-500M	56.75		
			SmolVLM-256M	47.17		
Reference	Finetuning	None	SmolVLM-500M	56.96		
			SmolVLM-256M	49.83		
Same-family distillation baselines						
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	59.00		
			SmolVLM-256M	50.88		
	Reverse KL	SmolVLM2-2.2B	SmolVLM-500M	44.73		
			SmolVLM-256M	29.75		
	Jensen-Shannon	SmolVLM2-2.2B	SmolVLM-500M	42.33		
			SmolVLM-256M	33.84		
Different-family single-teacher baselines						
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	61.49		
			SmolVLM-256M	53.95		
Heterogeneous multi-teacher baselines						
Multi-teacher	Uniform	Qwen2-VL-2B-Instruct	SmolVLM-500M	63.88		
		Qwen2.5-VL-3B-Instruct	SmolVLM-256M	54.57		
	SHM-CKA	Qwen2-VL-2B-Instruct	SmolVLM-500M	60.47		
		Qwen2.5-VL-3B-Instruct				
		InternVL3-1B			SmolVLM-256M	55.93
		Gemma-3-4B-it				
	Reinforced selection	Qwen2-VL-2B-Instruct	SmolVLM-500M	63.26		
		Qwen2.5-VL-3B-Instruct				
		InternVL3-1B			SmolVLM-256M	56.31
		Gemma-3-4B-it				
	GRACE	Qwen2-VL-2B-Instruct	SmolVLM-500M	69.38		
		Qwen2.5-VL-3B-Instruct				
		InternVL3-1B			SmolVLM-256M	60.73
		Gemma-3-4B-it				

Table 5.3: TextVQA results for the original students, fine-tuning, single-teacher baselines and heterogeneous multi-teacher methods. The table is organized by method family so the gain from adaptive weighting can be compared against simpler baselines.

Category	Method	Teacher	Student	DocVQA		
Non-distillation references						
Original	None	None	SmolVLM-500M	59.52		
			SmolVLM-256M	48.64		
Reference	Finetuning	None	SmolVLM-500M	61.96		
			SmolVLM-256M	42.03		
Same-family distillation baselines						
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	63.55		
			SmolVLM-256M	49.46		
	Reverse KL	SmolVLM2-2.2B	SmolVLM-500M	29.28		
			SmolVLM-256M	23.28		
	Jensen-Shannon	SmolVLM2-2.2B	SmolVLM-500M	36.43		
			SmolVLM-256M	33.12		
Different-family single-teacher baselines						
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	65.28		
			SmolVLM-256M	52.24		
Heterogeneous multi-teacher baselines						
Multi-teacher	Uniform	Qwen2-VL-2B-Instruct	SmolVLM-500M	69.59		
		Qwen2.5-VL-3B-Instruct	SmolVLM-256M	55.70		
	SHM-CKA	Qwen2-VL-2B-Instruct	SmolVLM-500M	63.32		
		Qwen2.5-VL-3B-Instruct				
		InternVL3-1B			SmolVLM-256M	57.24
		Gemma-3-4B-it				
	Reinforced selection	Qwen2-VL-2B-Instruct	SmolVLM-500M	64.00		
		Qwen2.5-VL-3B-Instruct				
		InternVL3-1B			SmolVLM-256M	58.53
		Gemma-3-4B-it				
	GRACE	Qwen2-VL-2B-Instruct	SmolVLM-500M	73.11		
		Qwen2.5-VL-3B-Instruct				
		InternVL3-1B			SmolVLM-256M	60.48
		Gemma-3-4B-it				

Table 5.4: DocVQA results for the original students, fine-tuning, single-teacher baselines and heterogeneous multi-teacher methods. The rows use the same grouping as Table 5.3 so the cross-benchmark pattern is easy to compare.

Category	Method	Teacher	Student	ChartQA
Non-distillation references				
Original	None	None	SmolVLM-500M	62.80
			SmolVLM-256M	45.04
Reference	Finetuning	None	SmolVLM-500M	63.00
			SmolVLM-256M	49.68
Same-family distillation baselines				
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	64.52
			SmolVLM-256M	56.80
	Reverse KL	SmolVLM2-2.2B	SmolVLM-500M	30.16
			SmolVLM-256M	17.20
	Jensen-Shannon	SmolVLM2-2.2B	SmolVLM-500M	43.75
			SmolVLM-256M	50.93
Different-family single-teacher baselines				
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	66.92
			SmolVLM-256M	60.96
Heterogeneous multi-teacher baselines				
Multi-teacher	Uniform	Qwen2-VL-2B-Instruct	SmolVLM-500M	75.60
		Qwen2.5-VL-3B-Instruct	SmolVLM-256M	63.68
	SHM-CKA	Qwen2-VL-2B-Instruct	SmolVLM-500M	76.88
		Qwen2.5-VL-3B-Instruct		
		InternVL3-1B		
		Gemma-3-4B-it		
	Reinforced selection	Qwen2-VL-2B-Instruct	SmolVLM-500M	70.27
		Qwen2.5-VL-3B-Instruct		
		InternVL3-1B		
		Gemma-3-4B-it		
	GRACE	Qwen2-VL-2B-Instruct	SmolVLM-500M	80.20
		Qwen2.5-VL-3B-Instruct		
		InternVL3-1B		
		Gemma-3-4B-it		

Table 5.5: ChartQA results for the original students, fine-tuning, single-teacher baselines and heterogeneous multi-teacher methods. This table uses the same method layout as the TextVQA and DocVQA comparisons.

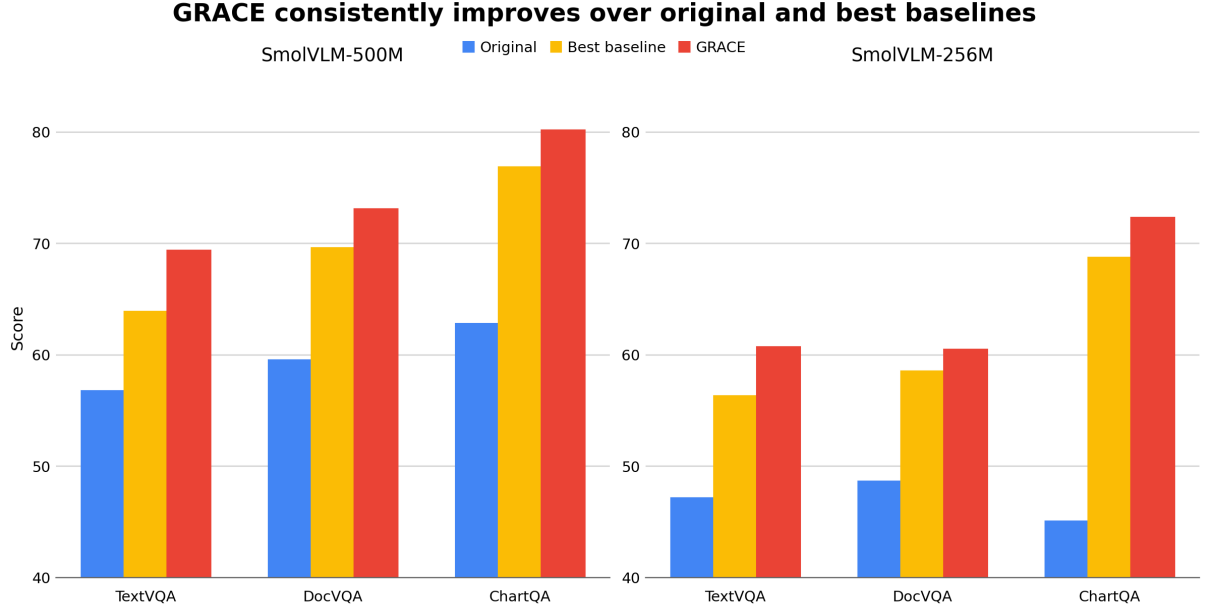


Figure 5.1: Grouped comparison of the original student, the strongest non-GRACE baseline and GRACE for both student sizes. The chart makes the pattern in Tables 5.3 to 5.5 easier to scan: GRACE stays above the best competing baseline on every benchmark.

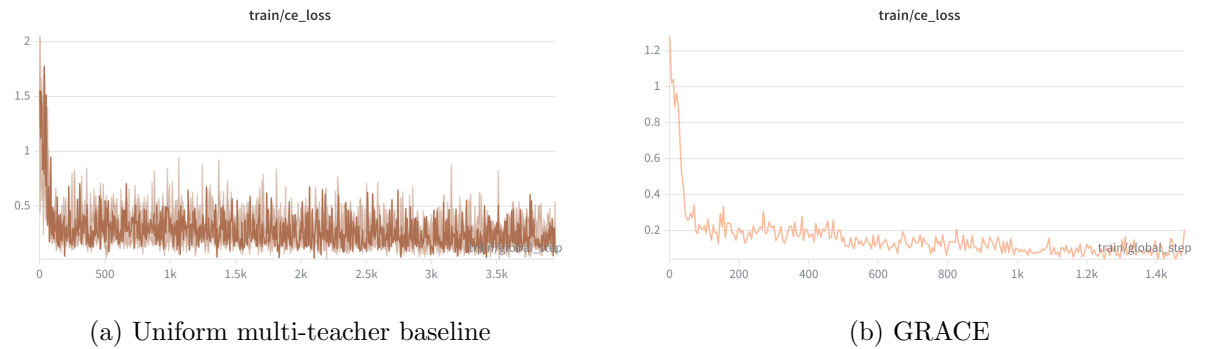
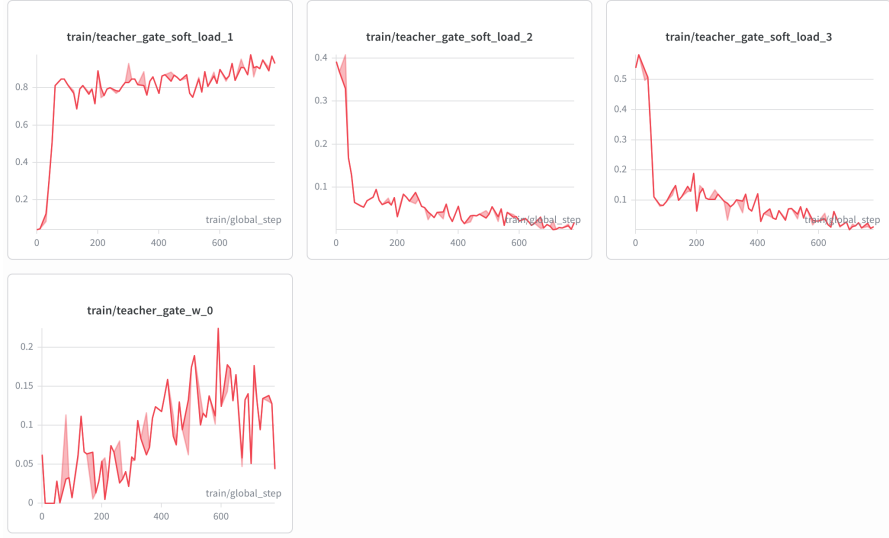
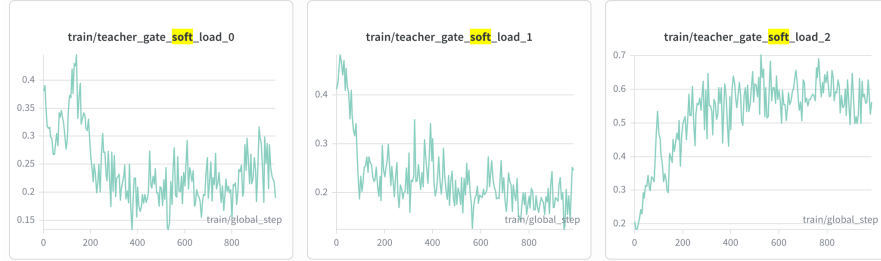


Figure 5.2: Representative training curves from a uniform multi-teacher run and a GRACE run. The GRACE trace is visibly smoother and shows smaller oscillations, which is consistent with the stability claim, but the figure is illustrative rather than a controlled matched-run comparison.



(a) Without GRACE refinement



(b) With GRACE

Figure 5.3: Representative routed-teacher load plots from one collapse case and one stable GRACE case. The figure is diagnostic, not a controlled matched-run ablation: it shows that the router can collapse toward one teacher, while a GRACE run with the current trie-Wasserstein loss keeps routing mass distributed across the teacher pool.

Benchmark	vs Original	vs Finetuning	vs Uniform	vs Best non-GRACE
TextVQA	+12.63 (+22.26%)	+12.42 (+21.80%)	+5.50 (+8.61%)	+5.38 (+8.41%)
DocVQA	+13.59 (+22.83%)	+11.15 (+18.00%)	+3.52 (+5.06%)	+3.52 (+5.06%)
ChartQA	+17.40 (+27.71%)	+17.20 (+27.30%)	+4.60 (+6.08%)	+3.32 (+4.32%)

Table 5.6: Absolute and relative gains of GRACE on SmolVLM-500M against the original model, direct finetuning, uniform weighting and the strongest non-GRACE baseline. Relative gain is computed as $(\text{GRACE} - \text{baseline})/\text{baseline}$.

representative GRACE run, the final load remains spread across teachers, so the student keeps the benefit of the heterogeneous pool.

Table 5.6 makes the same point numerically for the SmolVLM-500M student. Relative gain is computed as $(\text{GRACE} - \text{baseline})/\text{baseline}$.

- **The gains over the original student are large:** +22.26% on TextVQA, +22.83% on DocVQA and +27.71% on ChartQA. The gains over direct finetuning are almost as large: +21.80%, +18.00% and +27.30%, respectively.
- the gain over static uniform averaging is smaller, but it remains consistent across all three tasks. **This isolates the value of adaptive teacher weighting from the value of simply using more than one teacher.**
- **The same pattern holds for SmolVLM-256M.** GRACE rises from 47.17 to 60.73 on TextVQA (+28.75%), from 48.64 to 60.48 on DocVQA (+24.34%) and from 45.04 to 72.32 on ChartQA (+60.57%).
- it also stays ahead of direct finetuning and uniform weighting. Relative to the uniform baseline, the gains are +11.29% on TextVQA, +8.58% on DocVQA and +13.57% on ChartQA.

The same trend appears if the reference point is the strongest teacher in the GRACE pool. Let $g_{\text{orig}} = y_T - y_{\text{orig}}$ and $g_{\text{grace}} = y_T - y_{\text{grace}}$. The fraction of the gap closed is $1 - g_{\text{grace}}/g_{\text{orig}}$.

The gain is not just a small increment over a weak baseline. On two benchmarks, GRACE removes about half of the original teacher-student gap. Even on DocVQA, where the teacher ceiling is highest, it still closes almost 40% of the gap.

The student can even outperform some individual teachers in the pool. On TextVQA, GRACE reaches 69.38 and exceeds Gemma-3-4B-it at 57.80 by +11.58 points

Benchmark	Best teacher	Original gap	GRACE gap	Gap closed
TextVQA	79.70	22.95	10.32	55.03%
DocVQA	93.90	34.38	20.79	39.53%
ChartQA	94.24	31.44	14.04	55.34%

Table 5.7: Gap-to-teacher analysis for SmolVLM-500M under GRACE. The teacher reference is the strongest model in the four-teacher pool for each benchmark and the last column reports the fraction of the original gap that GRACE closes.

(+20.03%). On ChartQA, it exceeds InternVL3-1B at 75.30 by +4.90 points (+6.51%) and Gemma-3-4B-it at 63.60 by +16.60 points (+26.10%). This shows that the final student is not simply bounded by the weaker teachers it learns from.

One negative result is also consistent: reverse KL is much weaker than forward KL. This suggests that the more mode-seeking objective is a poor fit for these compact multimodal students.

SHM-CKA is not enough by itself. SHM-CKA uses the same heterogeneous teacher pool, but replaces routed weighting with hidden-state matching. It is competitive on ChartQA, but clearly weaker on TextVQA and DocVQA. This implies that representation alignment alone is not sufficient. Teacher assignment matters, especially when the pool contains teachers with different strengths and tokenizers.

A clear limitation remains on the vision side. The current method distills supervised answer-token logits, pools student hidden states for routing and scores teachers from CE/KD agreement at the output side. This is effective, but it does not explicitly align visual features, cross-attention patterns, or region-level evidence inside the VLM. The method therefore treats the vision encoder mostly through its effect on the final answer distribution, not as a first-class distillation target.

5.2 Representation Similarity Analysis

We use Centered Kernel Alignment (CKA) as an analysis tool to examine whether teacher and student layers exhibit clear correspondence. For centered hidden-state matrices $\tilde{\mathbf{H}}_s$ and $\tilde{\mathbf{H}}_t$, linear CKA is

$$\text{CKA}(\tilde{\mathbf{H}}_s, \tilde{\mathbf{H}}_t) = \frac{\left\| \tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_s \right\|_F^2}{\left\| \tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_t \right\|_F \left\| \tilde{\mathbf{H}}_s^\top \tilde{\mathbf{H}}_s \right\|_F}. \quad (5.1)$$

Values closer to 1 indicate stronger representation similarity [65].

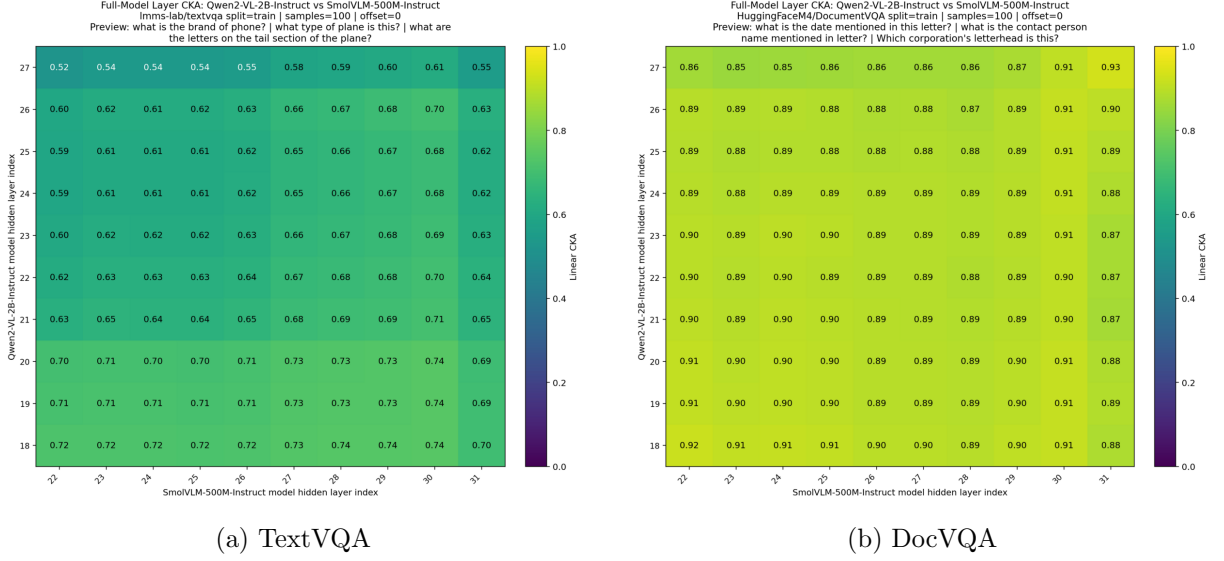


Figure 5.4: Final-layer CKA heatmaps from our experiments for the teacher-student pair Qwen2-VL-2B and SmolVLM-500M on TextVQA and DocVQA. Brighter regions and values closer to 1 indicate stronger representation similarity, but they do not imply that direct layer matching is the best training signal.

The main question is whether strong internal similarity is enough to justify layer-wise distillation. Our results suggest that it is not, as shown in the figures and tables below.

The final-layer maps already show that the teacher and student are not arbitrarily far apart in representation space.

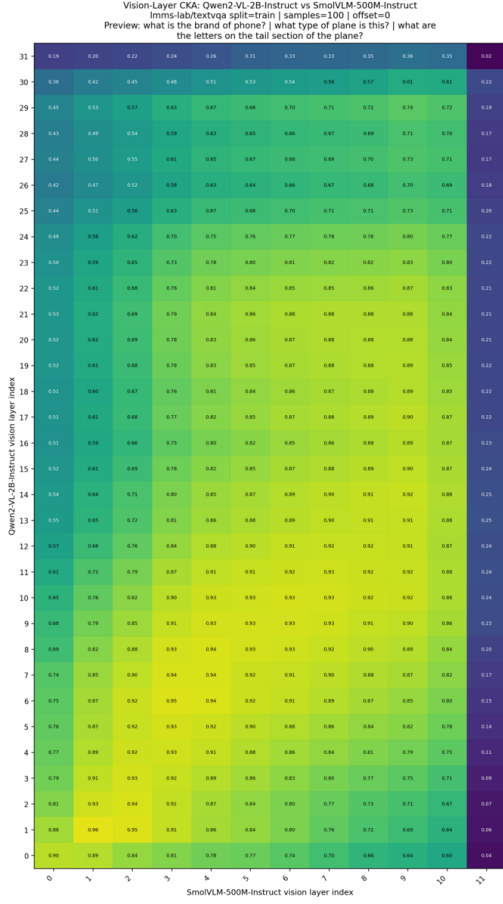
Figures 5.4 and 5.5 show that the teacher and student can be relatively close in both the language and vision stacks.

However, Tables 5.3, 5.4, and 5.5 still show a clear benchmark gap between compact students and stronger models. The layer-wise baseline also fails to consistently outperform output-level distillation.

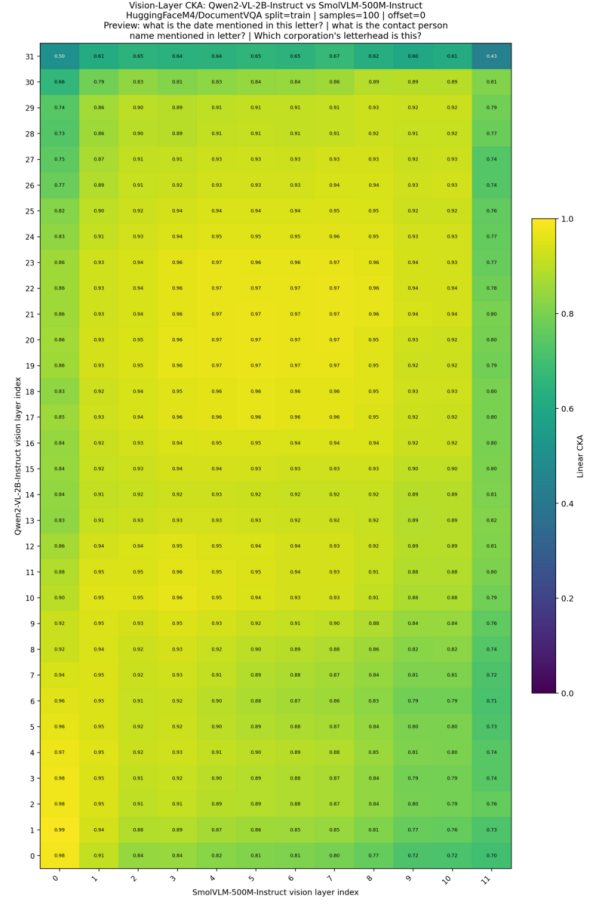
This supports the design choice in GRACE. High hidden-state similarity alone is not a reliable signal that direct layer matching will produce better task transfer.

5.3 Ablation Study

We report a compact teacher-assignment study on SmolVLM-500M. The goal is to test whether the gain comes only from using more teachers, or from the specific adaptive weighting rule in GRACE.



(a) TextVQA



(b) DocVQA

Figure 5.5: Vision-layer CKA heatmaps for the same teacher-student pair on TextVQA and DocVQA. Again, brighter regions and values closer to 1 indicate stronger similarity, while the benchmark gaps show that similar internal structure is still not enough by itself.

Variant	TextVQA	DocVQA	ChartQA
ULD (single cross-family teacher)	61.49	65.28	66.92
Uniform multi-teacher	63.88	69.59	75.60
SHM-CKA	60.47	63.32	76.88
Reinforced selection	63.26	64.00	70.27
GRACE	69.38	73.11	80.20

Table 5.8: Teacher-assignment ablation on SmolVLM-500M across TextVQA, DocVQA and ChartQA. The variants isolate single-teacher transfer, static multi-teacher averaging and adaptive multi-teacher weighting under the same student size.

Student model	Scale	Original	Best baseline	GRACE
SmolVLM-256M	256M	48.64	58.53	60.48
LFM2.5-VL-450M	450M	N/A	N/A	N/A
SmolVLM-500M	500M	59.52	69.59	73.11

Table 5.9: Partial student-scaling comparison on DocVQA. The table reports the original model, the strongest non-GRACE baseline and GRACE for each student size, while the LFM2.5-VL-450M row remains open for future runs.

Table 5.8 suggests that heterogeneous supervision is useful, but static or partially adaptive alternatives still miss part of the gain.

We also include a student-scaling study on DocVQA. The goal is to check whether the same training recipe remains effective as the student capacity changes from SmolVLM-256M to LFM2.5-VL-450M and SmolVLM-500M.

Table 5.9 gives the currently available scaling comparison and leaves the LFM row for future completion.

5.4 Qualitative Analysis

This section presents representative examples to inspect how distillation affects text reading and basic reasoning behavior.

5.5 Discussion

- **The results show a clear ordering.** Same-family KL baselines are weakest, single cross-family distillation is stronger, uniform weighting improves again and GRACE is best overall [11, 12]. This means that the problem is not solved by one trick alone. Heterogeneous teachers provide better supervision, but the student still needs a cross-tokenizer loss, sample-dependent routing and a stable post-routing weighting rule.
- the code-level implementation helps explain that pattern. The current training recipe uses a shared-trie Wasserstein loss for mismatched tokenizers, a routed multi-teacher setup with hard-routing warmup, top- k capacity control and GRACE score refinement with EMA-smoothed score updates [12, 14, 43]. The smoother training curve in Figure 5.2 is consistent with that design: the router narrows the teacher

set first, then GRACE reweights only the routed teachers instead of averaging all teachers equally.

- the largest gain appears on ChartQA and the same trend also holds for the smaller 256M student. This suggests that sample-dependent teacher assignment is most useful when the task mixes visual parsing and numerical reasoning. It also shows that the method is not only helping the larger compact student. The improvement persists in the more resource-constrained regime.
- the comparison with SHM-CKA is also informative. SHM-CKA uses the same heterogeneous teacher pool, but it is less consistent than GRACE. Together with the CKA heatmaps, this suggests that internal similarity alone is not enough. **Good transfer still depends on which teacher is emphasized for each sample** [33, 65].

Two limitations remain:

- the reported results are single-run numbers, because offline teacher-logit generation, remote storage and multi-teacher training are expensive. In our setup, a typical H200 run takes about 7 hours and costs about 21 USD on Vast.ai before cache-generation cost. This budget is the main reason some broader sweeps are still missing and are deferred to later work.
- the current implementation still depends on large remote teacher-logit caches. The method is effective, but the systems cost is still significant.

Conclusion and Future Work

This thesis addressed compact visual language model distillation under heterogeneous teachers. The student is much smaller than the teacher pool, the teachers do not help every sample equally and their tokenizers do not match. This thesis presents GRACE, a routed multi-teacher framework with trie-Wasserstein cross-tokenizer distillation, capacity-aware selection and EMA-smoothed GRACE weighting. The code follows the same order: route teachers first, refine their weights with GRACE scores, then aggregate the routed trie-Wasserstein KD loss [12, 14, 43].

Empirically, GRACE is the strongest method in the main comparison. For SmolVLM-500M, it reaches 69.38 on TextVQA, 73.11 on DocVQA and 80.20 on ChartQA. For SmolVLM-256M, it reaches 60.73, 60.48 and 72.32. These results improve over the original students, direct finetuning and the best competing baselines in Chapter 5. On SmolVLM-500M, GRACE closes about 55% of the original teacher-student gap on TextVQA and ChartQA and about 40% on DocVQA. Taken together, these results establish that selective routing plus post-routing score refinement is a strong recipe for compact text-rich VLM distillation.

Future Work

- stronger evaluation. Repeated runs would make the comparison more rigorous, especially for the smaller gaps between the stronger multi-teacher baselines.
- better systems efficiency. The current pipeline still relies on large offline teacher-logit caches, so reducing storage and I/O cost would make the method easier to scale to more datasets and larger teacher pools.
- broader method extensions. The current recipe is still answer-token centric, so one important next step is explicit visual distillation: visual-token alignment, cross-attention transfer, or region-level matching. The method can also study richer router inputs, stronger objective-conflict handling and teacher pools that mix specialized document, chart and general VLM experts [40–42].

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